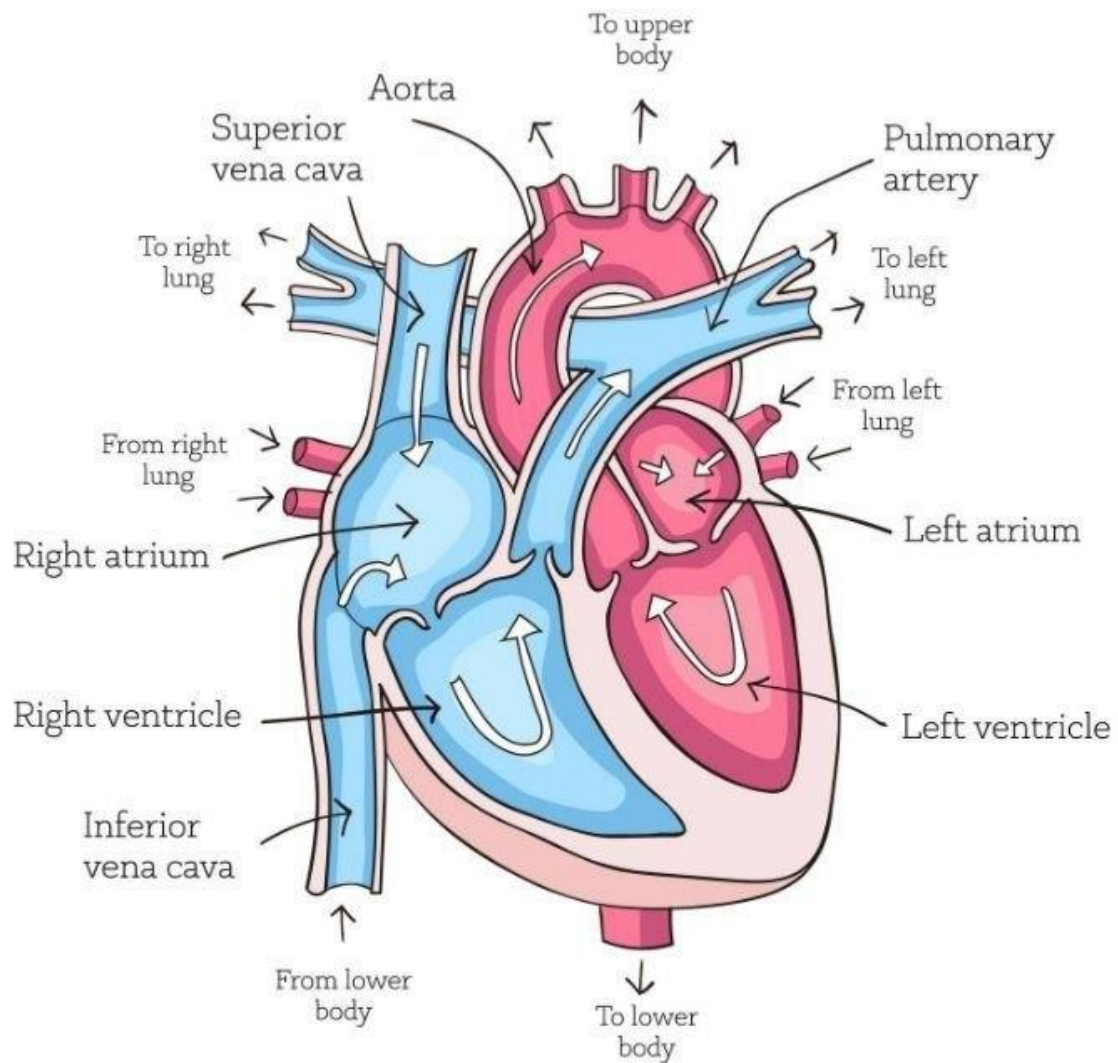


[25/06/2026 11:23 AM] hssnd: Welcome to anatomy of the Cardiovascular System ☺

Today we are going to cover:

- ⚡ Introduction to CVS
 - ⚡ Thoracic cavity
 - ⚡ Shape size and position of the heart
-



THE HUMAN HEART

[25/06/2026 11:24 AM] hssnd: Dr. Doc:

□ Cardiovascular System & Thoracic Mediastinum Anatomy

I. Introduction to the Cardiovascular System

🔍 The cardiovascular system (circulatory system) is the organ system responsible for transporting oxygen, nutrients, hormones, and waste products to and from body cells. This continuous transport helps maintain homeostasis within the body.

✳️ Primary Components of the Cardiovascular System

1. □ The Heart

The heart is a muscular pump that maintains blood circulation through rhythmic and repeated contractions.

◆ It consists of four chambers:

Right atrium – receiving chamber

Left atrium – receiving chamber

Right ventricle – pumping chamber

Left ventricle – pumping chamber

2. □ Blood Vessels

Blood vessels form a transport network throughout the body.

● Arteries

Carry blood away from the heart

● Veins

Carry blood toward the heart

○ Capillaries

Smallest blood vessels

● Site of exchange of: Oxygen, Nutrients, Waste products, and Hormones.

3. □ Blood

Blood is the fluid medium that transports substances throughout the body.

🔄 Major Circulatory Routes

1. 🦷 Pulmonary Circuit

Facilitated mainly by the right side of the heart.

■ Pathway:

- Deoxygenated blood leaves the right ventricle
- Travels through the pulmonary trunk to the lungs
- Gas exchange occurs in the lungs
- Oxygenated blood returns to the left side of the heart

**Function:

Oxygenates blood

Removes carbon dioxide

2. 🌐 Systemic Circuit

Facilitated mainly by the left side of the heart.

■ Pathway:

- Oxygenated blood leaves the left ventricle
- Passes through the aorta
- Distributed to all body organs (including the heart wall itself)
- Deoxygenated blood returns through:

•Superior vena cava

•Inferior vena cava

**Function:

Supplies oxygen and nutrients to body tissues including other parts of the lung and the wall of the heart.

□ II. The Thoracic Cavity

The thoracic cavity is the space within the chest located:

☆Deep to the thoracic wall

☆Superior to the diaphragm

☆Inferior to the root of the neck

It contains organs and tissues belonging to:

✓ Respiratory system

✓ Cardiovascular system

✓ Nervous system

✓ Immune system

✓ Digestive system

🔑 Boundaries of the Thoracic Cavity

⬆ Superior Boundary

Superior thoracic aperture also called "the root of the neck"

⬇ Inferior Boundary

Diaphragm

⬅ ➡ Lateral Boundaries

Thoracic wall and ribs

● Anterior Boundary

Sternum

Anterior thoracic wall

⬅ BACK Posterior Boundary

Bodies of the thoracic vertebrae

☐ Compartments of the Thoracic Cavity

The thoracic cavity is divided into three major compartments:

1. ☐ Right Pleural Cavity

Contains:

Right lung

Pleural membranes

2. ☐ Left Pleural Cavity

Contains:

Left lung

Pleural membranes

3. □ Mediastinum

The mediastinum is the central compartment located between the two pleural cavities.

Contains:

All thoracic viscera except the lungs, including:

- Heart
- Great vessels
- Trachea
- Esophagus
- Nerves
- Lymphatics

□ III. The Mediastinum

The mediastinum extends:

From the superior thoracic aperture to the diaphragm

From the sternum to the thoracic vertebrae

✂ Division of the Mediastinum

A horizontal plane passing through the:

☆ Sternal Angle (Angle of Louis)

Corresponds to:

Intervertebral disc between T4 and T5 vertebrae

This plane divides the mediastinum into:

↑ Superior Mediastinum

↓ Inferior Mediastinum

🏠 A. Superior Mediastinum

🔑 Boundaries

☆ Anteriorly

Manubrium of sternum

☆ Posteriorly

Bodies of T1–T4 vertebrae

☆ Inferiorly

Plane passing through the sternal angle

📖 Contents of the Superior Mediastinum

The contents are arranged from anterior to posterior in specific planes.

1. ☐ Glandular Plane

Contains:

✳️ Thymus

Most prominent in children

Lies immediately posterior to the manubrium

2. ☐ Venous Plane

Contains:

✓ Right brachiocephalic vein

✓ Left brachiocephalic vein

✓ Superior vena cava (SVC)

3. ● Arterial Plane

Contains:

✳️ Arch of the Aorta

Its major branches are:

✓ Brachiocephalic trunk

[25/06/2026 11:24 AM] hssnd: ✓ Left common carotid artery

✓ Left subclavian artery

4. ⚡ Nervous Plane

Contains:

✓ Vagus nerves (CN X)

✓ Phrenic nerves (C3–C5)

✓ Cardiac plexus

□ The phrenic nerve is the major motor nerve of the diaphragm.

5. 🦷🦷 Visceral Plane

Contains:

✓ Trachea: Located anteriorly

✓ 🦷 Esophagus: Located posteriorly

6. 💧 Lymphatic Plane

Contains:

✓ Thoracic duct

✓ Lymphatic trunks

□ B. Inferior Mediastinum

The inferior mediastinum is subdivided by the pericardial sac into:

1. Anterior mediastinum

2. Middle mediastinum

3. Posterior mediastinum

1. □ Anterior Mediastinum

📍 Location

Between the body of the sternum and the pericardium

 Smallest subdivision of the mediastinum

 Contents

- Loose connective tissue
- Sternopericardial ligaments
- Fat
- Lymphatic vessels
- Lymph nodes
- Branches of internal thoracic vessels
- Inferior part of thymus (especially in children)

2. Middle Mediastinum

 Location

Central part of the inferior mediastinum.

Boundaries: Anteriorly and posteriorly by the pericardium

 Contents

 Heart: Main organ of the middle mediastinum

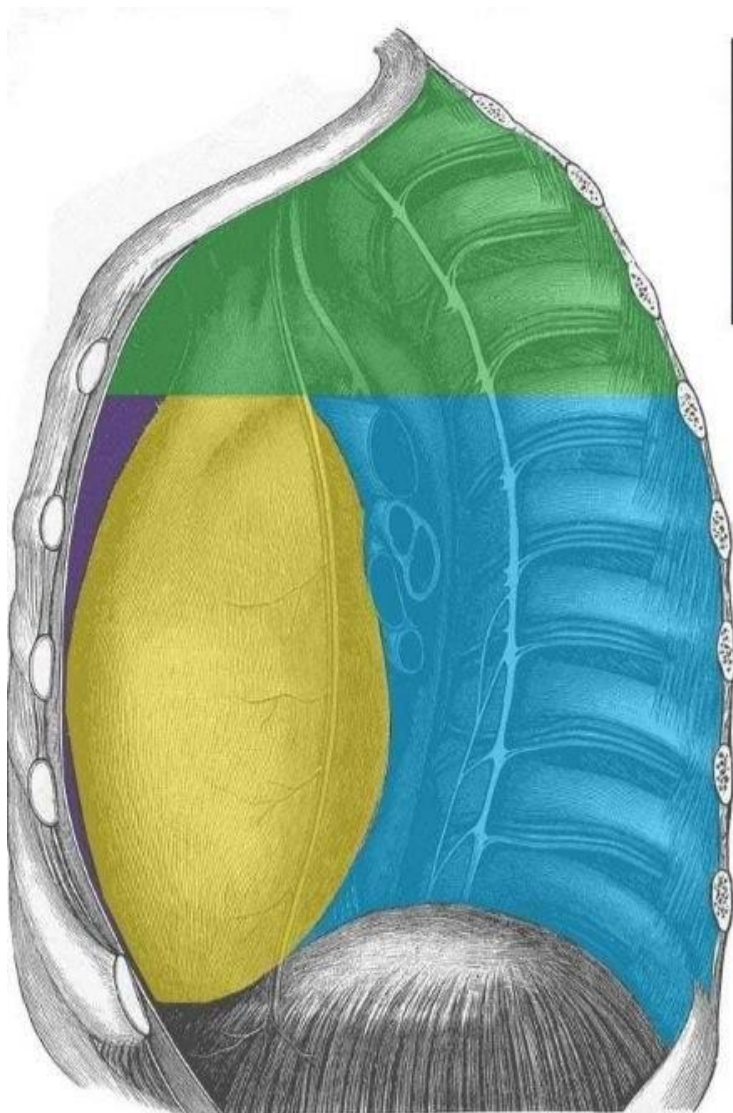
 Pericardium: Fibroserous sac surrounding the heart

 Phrenic Nerves

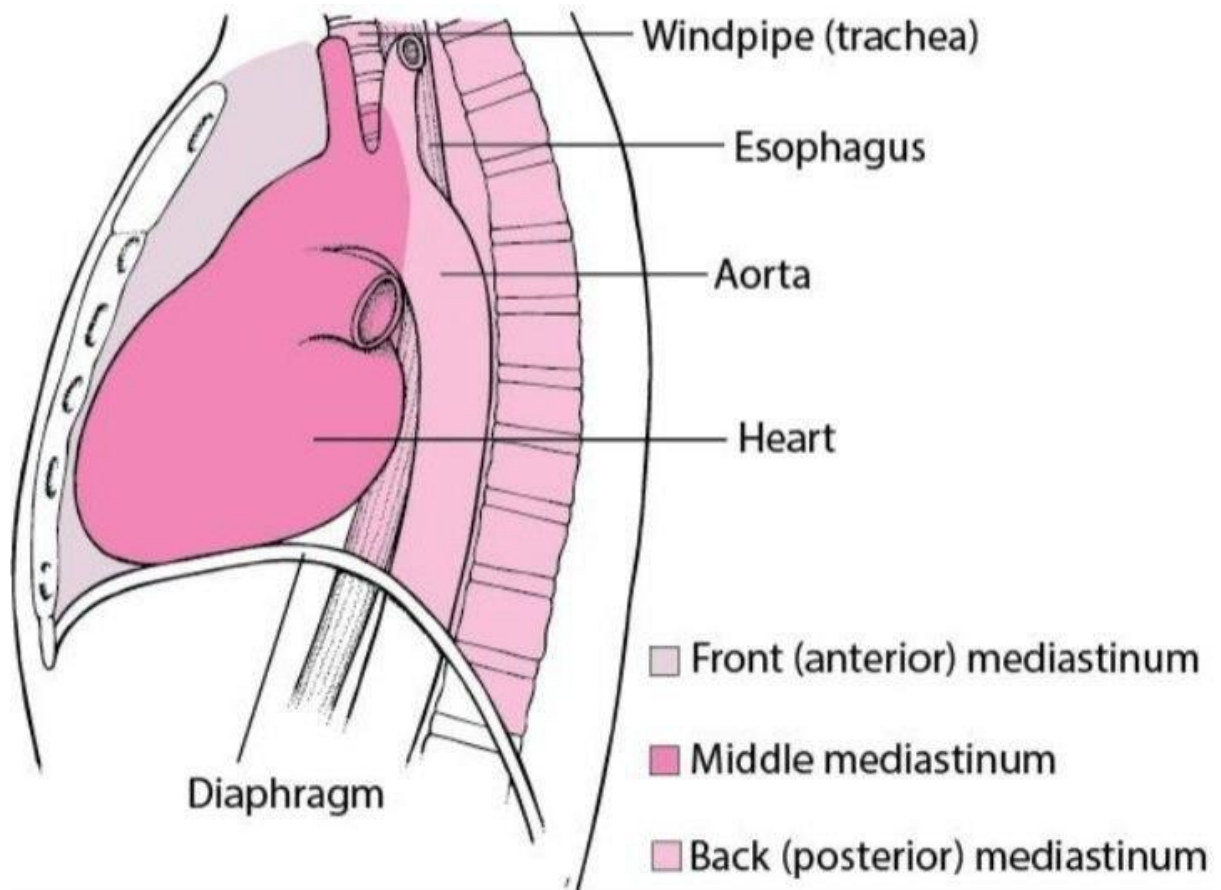
 Roots of Great Vessels

Including:

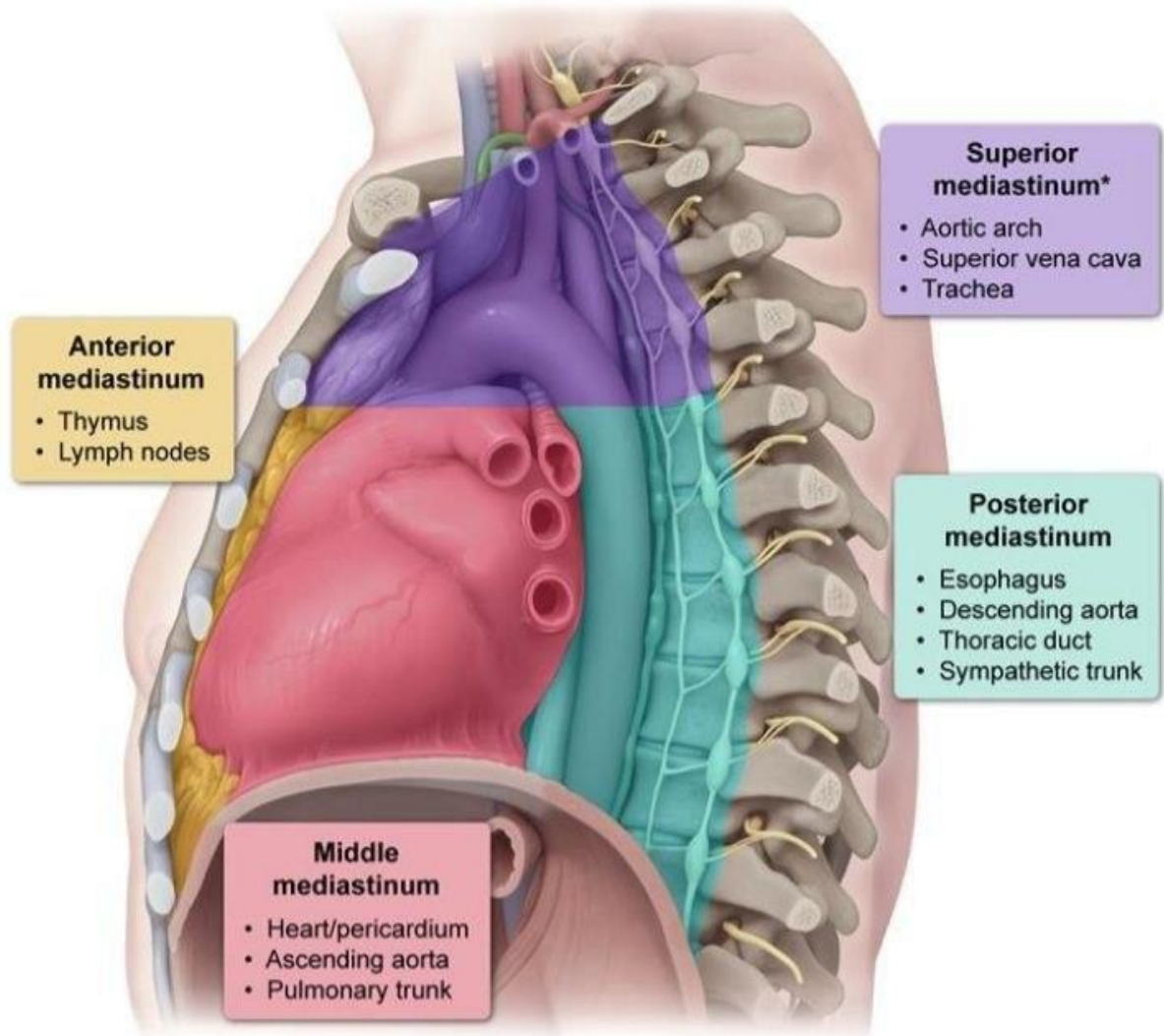
- Ascending aorta
- Pulmonary trunk
- Superior vena cava
- Inferior vena cava



- Superior mediastinum
- Anterior mediastinum
- Middle mediastinum
- Posterior mediastinum



Mediastinal compartments



*Also contains upper portion of esophagus, thoracic duct & sympathetic trunk.

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[25/06/2026 11:27 PM] hssnd: ♥ Position, Size, and Shape of the Heart

📍 Position of the Heart

The heart lies within the: Middle Mediastinum

☐ It is situated:

Between the two lungs

Posterior to the sternum

Anterior to the vertebral column

Orientation of the Heart

The heart lies obliquely within the thorax.

☐ Base

Directed: Posteriorly, Slightly superiorly

▲ Apex

Directed: Anteriorly, Inferiorly, To the left

Midline Relationship

Approximately two-thirds of the heart lies to the left of the sternum midline

Anatomical Landmarks

▲ Apex

Located deep to: Left 5th intercostal space

☐ Base

Extends to the level of: 2nd intercostal space.
This is where the great vessels attach.

↔ Deflection

Right side of heart → deflected anteriorly

Left side of heart → deflected posteriorly

▲ Shape of the Heart

The heart is:

 Pyramidal in shape

Features:

○ Broad superior base

○ Pointed inferior apex

□ External Grooves (Sulci)

The surface contains grooves called sulci that mark chamber boundaries and contain coronary vessels.

Major Sulci:

- Coronary sulcus
- Anterior interventricular sulcus
- Posterior interventricular sulcus

☞ Comparison

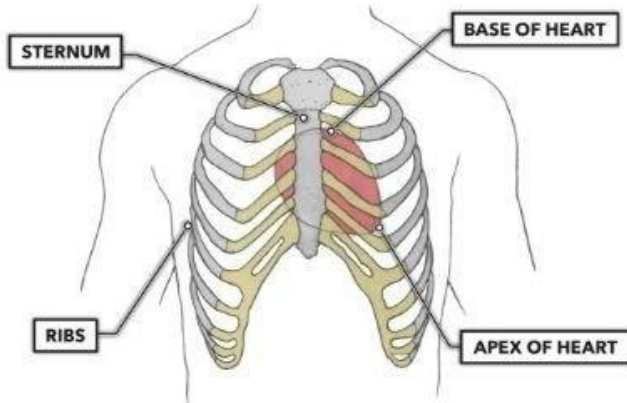
The heart is approximately: “The size of a closed fist”

✎ Size of the Heart

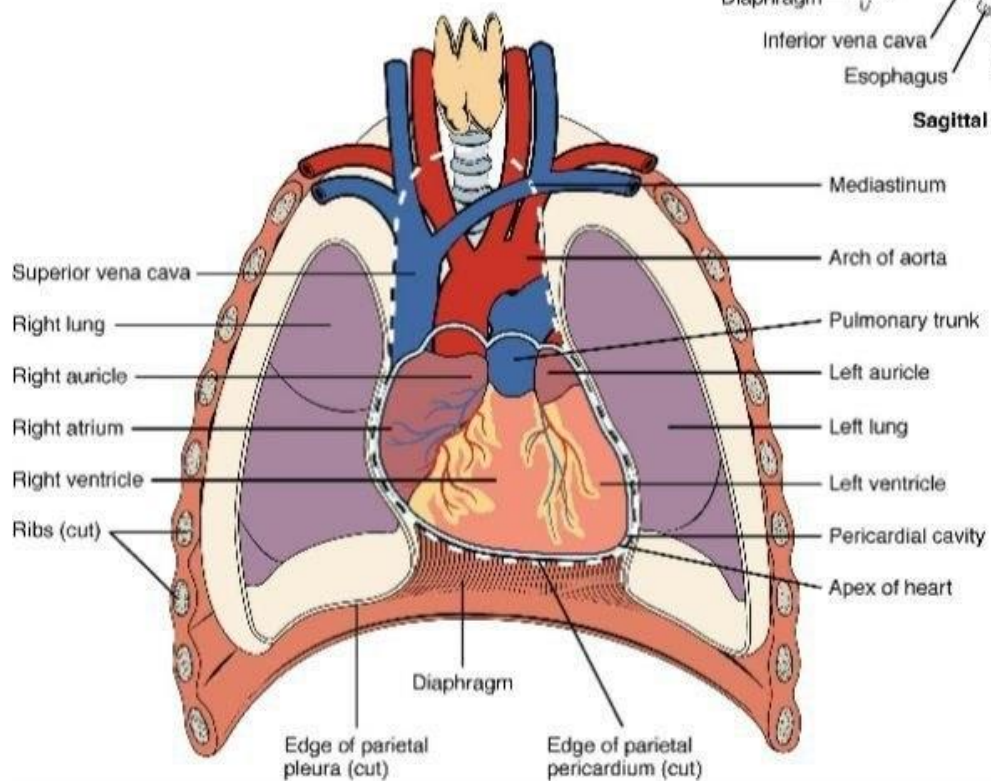
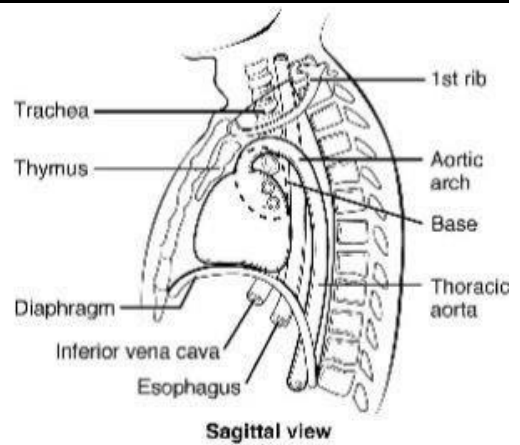
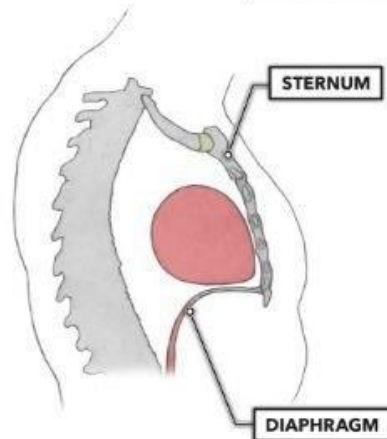
Average adult heart dimensions:

MeasurementApproximate Width9cm, Length13 cm, Thickness6 cm.

ANTERIOR VIEW



LATERAL VIEW



[25/06/2026 11:28 PM] hssnd: Dr. Doc:

♥ ANATOMY OF THE HEART CONTINUED

♥ I. Primary Functions of the Heart.

□ The heart is a hollow, four-chambered muscular organ that acts as two pumps in one.

□ The Right Side (Pulmonary Circulation) Receives oxygen-poor blood from the body and pumps it through the pulmonary trunk to the lungs for gas exchange.

✂ Flow:

Body → Right Atrium → Right Ventricle → Pulmonary Trunk → Lungs

🌐 The Left Side (Systemic Circulation) Receives oxygen-rich blood from the lungs via pulmonary veins and pumps it into the aorta for distribution to the entire body.

✂ Flow:

Lungs → Left Atrium → Left Ventricle → Aorta → Body

☆ Four Critical Functions of the Heart

1 Separation

▲ Keeps oxygen-poor and oxygen-rich blood in separate chambers.

2 Directional Flow

▲ Ensures blood flows in one direction through pulmonary and systemic circulations.

3 Pressure Generation

▲ Creates blood pressure necessary to move blood through vessels.

4 Regulation

▲ Adjusts blood supply according to metabolic needs of tissues.

🔑 II. Position and Orientation

Location

- ✓ The heart lies in the middle mediastinum of the thoracic cavity.
- ✓ Approximately 2/3 lies left of the midsternal line.
- ✓ Approximately 1/3 lies right of the midsternal line.

▲ Shape

☐ The heart is: Pyramidal ▲ or Trapezoidal

*The apex is the pointed inferior end of the heart. Formed Mainly By: ☐ Left ventricle

Direction:

- ☐ ↓ Downward
- ☐ → Forward
- ☐ ← To the left

☐ Surface Landmark:

📍 Left 5th intercostal space

📍 Just medial to the midclavicular line

*The base is the posterior aspect of the heart. Formed Mainly By: ☐ Left atrium + Small portion of right atrium

Direction:

☐ → Toward the right shoulder

Vertebral Level:

📍 T6–T9 vertebrae

Structures Opening Into Base:

☐ Four pulmonary veins

☐ ↑ Superior vena cava

☐ ↓ Inferior vena cava

III. Protective Covering: The Pericardium

The heart is enclosed within a double-walled fibro-serous sac called the pericardium.

☐ 1. Fibrous Pericardium

- ✓ A tough outer layer made of dense irregular connective tissue.

Functions

- ✓ Protects the heart
- ✓ Prevents overdistension
- ✓ Prevents overfilling
- ✓ Maintains position in mediastinum
- ✓ Keeps large vessels open
- ✓ Supports the serous pericardium

Attachments

↑ Superiorly

🔗 Continuous with the tunica adventitia of: Aorta, Pulmonary trunk, Great vessels.

→ Anteriorly

🔗 Attached to the sternum through:
📌 Sternopericardial ligaments

↓ Inferiorly

🔗 Fused with the central tendon of the diaphragm through:
📌 Pericardiophrenic ligament

💧 2. Serous Pericardium

🔗 A delicate double-layered membrane lined by mesothelium.

□ Parietal Layer

Fused to inner surface of fibrous pericardium.

♡ Visceral Layer (Epicardium)

Covers the surface of the heart. Reflects onto the roots of the great vessels.

🔗 Pericardial Cavity

□ Potential space between:

📌 Parietal layer

📌 Visceral layer

Contains:

□ Pericardial fluid

🧑 Function

Reduces friction during cardiac movements.

□ Blood Supply and Innervation of the Pericardium

☐ Fibrous and Parietal Pericardium

Arterial Supply

- ☐ Pericardiophrenic artery (main)
- ☐ Musculophrenic artery
- ☐ Bronchial arteries
- ☐ Esophageal arteries
- ☐ Superior phrenic arteries

Venous Drainage

- ☐ Pericardiophrenic veins
- ☐ Azygos veins

Nerve Supply

⚡ Phrenic nerve (C3–C5)

♡ Visceral Pericardium (Epicardium)

Blood Supply

- ☐ Coronary arteries

Innervation

⚡ Autonomic fibers

- ☐ Clinical Correlation

Pain from the fibrous and parietal pericardium is carried by:

⚡ Phrenic nerve (C3–C5)

Pain is commonly referred to:

👉 Shoulder region

☒ 3. Pericardial Sinuses

These are recesses formed during development when the primordial heart tube folds.

[25/06/2026 11:28 PM] hssnd: ↔ Transverse Pericardial Sinus:

Located:

← Posterior to:

Ascending aorta

Pulmonary trunk

→ Anterior to:

Superior vena cava

Atria

🏠 Surgical Importance

A ligature can be passed through this sinus to temporarily stop blood flow through:

- ☐ Ascending aorta
- ☐ Pulmonary trunk

during cardiac surgery.

🌀 Oblique Pericardial Sinus:

A blind recess located posterior to the base of the heart.

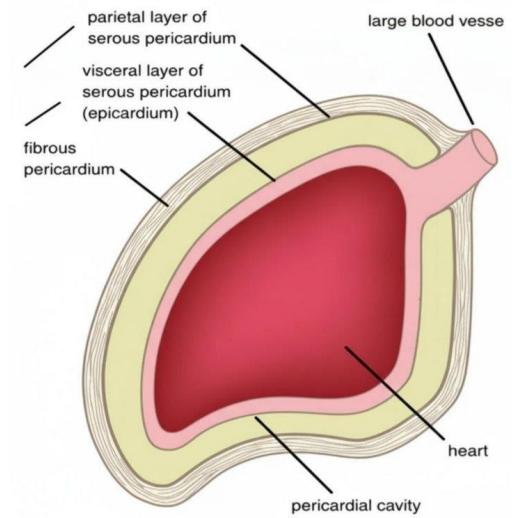
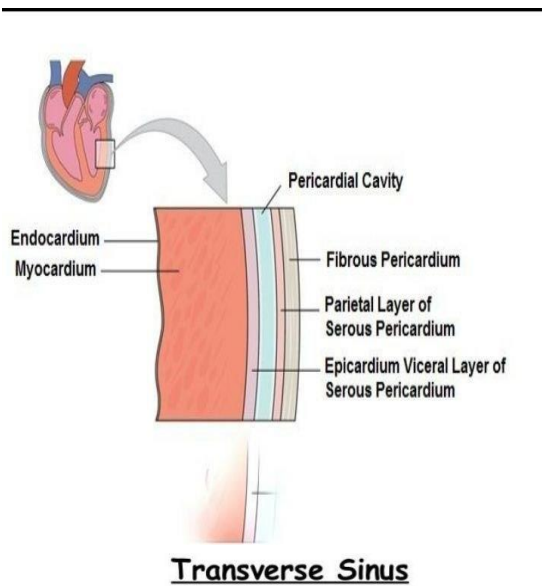
☐ Boundaries

Side Boundaries:

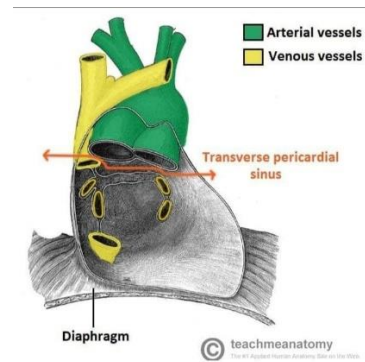
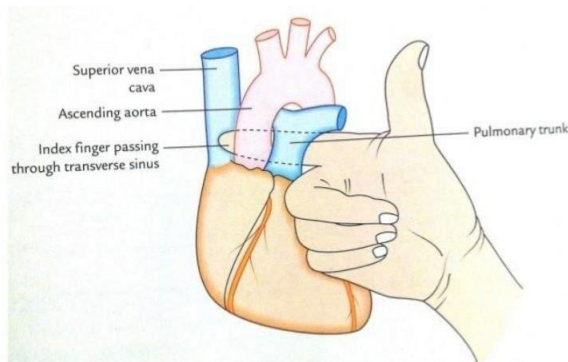
- ☐ Pulmonary veins
- ☒ Inferior vena cava

Posterior Boundary:

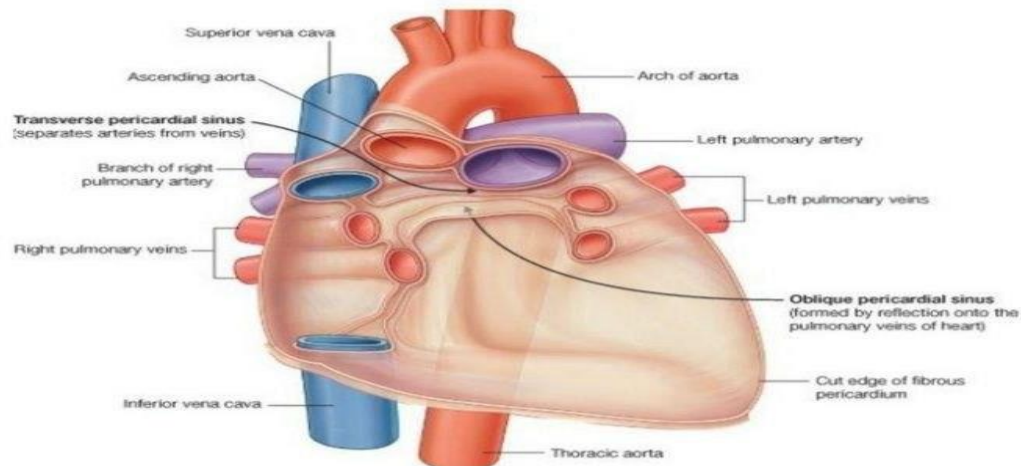
 Esophagus covered by pericardium



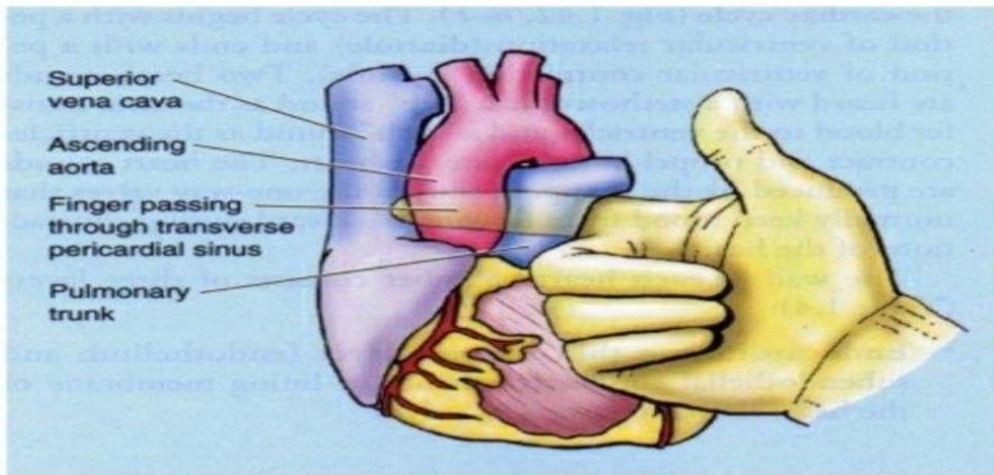
Transverse Sinus



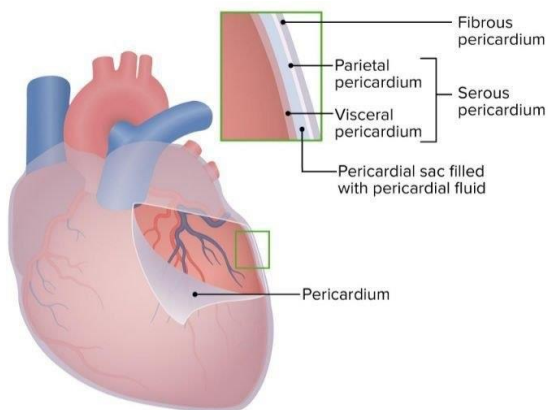
Transverse & Oblique sinus



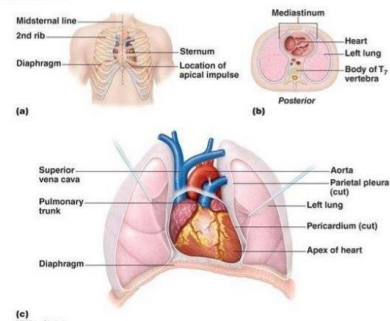
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Relations:
Aorta and Pulmonary trunk- anterior & SVC- posterior



The heart is located in the mediastinum, surrounded by a **pericardium**:



[25/06/2026 11:29 PM] hssnd:

□ IV. Layers of the Heart Wall

The wall of the heart consists of three layers.

1 Endocardium

Thin endothelial inner lining.

Covers chambers.

Covers heart valves.

2 Myocardium

Thick muscular middle layer.

Functions:

♥ Responsible for contraction

♥ Generates pumping force

3 Epicardium

Outer layer of heart wall.

Equivalent to:

♥ Visceral layer of serous pericardium

🔍 V. External Anatomy: Surfaces, Borders, and Sulci

☐ Surfaces of the Heart

☐ Sternocostal (Anterior) Surface

Formed mainly by:

♥ Right ventricle

☐ Diaphragmatic (Inferior) Surface

Formed mainly by:

♥ Left ventricle

and partly by:

♥ Right ventricle

☐ Right Pulmonary Surface

Formed by:

♥ Right atrium

☐ Left Pulmonary Surface

Formed by:

♥ Left ventricle

Produces the cardiac impression on the left lung.

Borders of the Heart

Right Border

Formed by:

♥ Right atrium

Extends between:

 SVC

 IVC

 Inferior Border

Formed mainly by:

♥ Right ventricle

and slightly by:

♥ Left ventricle

 Left Border

Formed mainly by:

♥ Left ventricle

and partly by:

👂 Left auricle

 Superior Border

Formed by:

♥ Right atrium

♥ Left atrium

👂 Auricles

From this border emerge:

☐ Ascending aorta

☐ Pulmonary trunk

Sulci (Grooves)

 Coronary (Atrioventricular) Sulcus: Encircles the heart near the base.

Function:

Separates: Atria, Ventricles

 Anterior Interventricular Sulcus: Separates the ventricles externally.

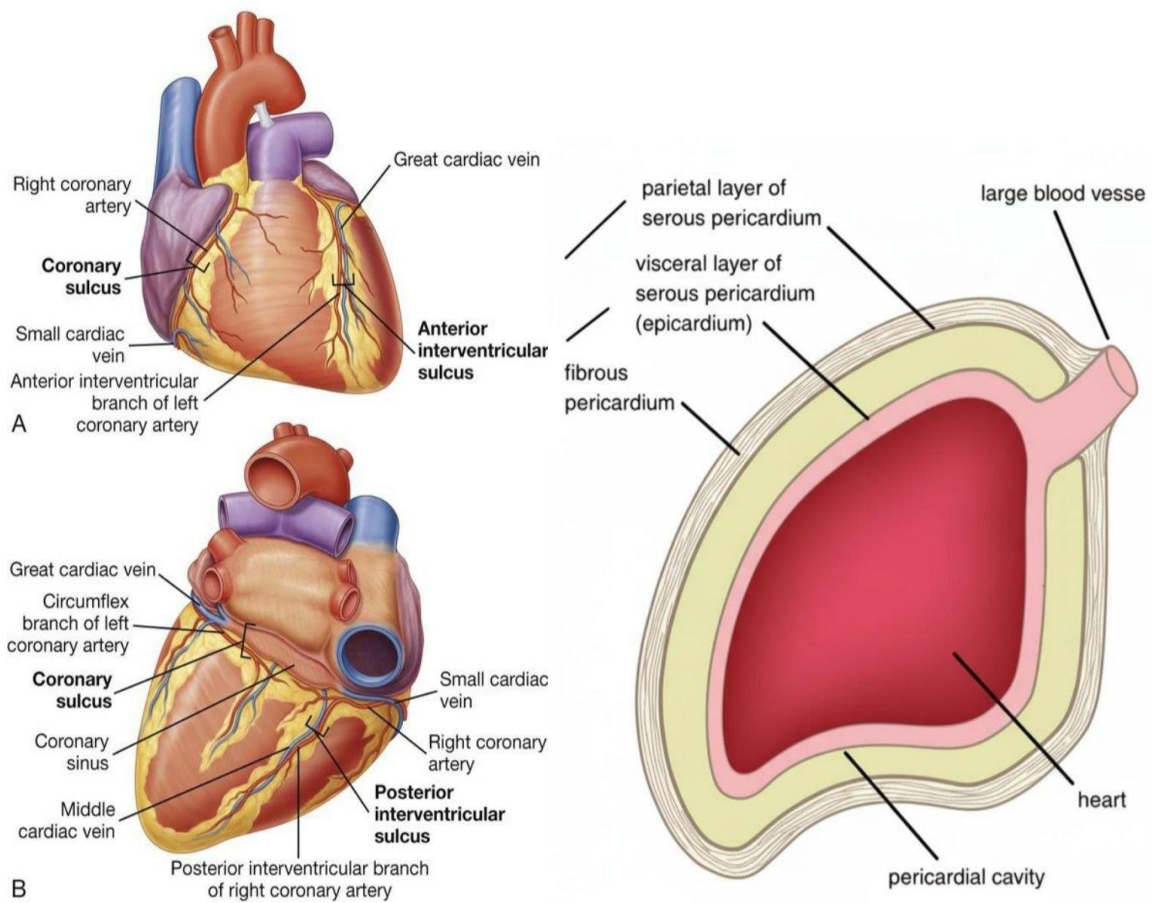
Contains:

☐ Coronary vessels

 Posterior Interventricular Sulcus: Separates the ventricles externally.

Contains:

- ☐ Coronary vessels
- ☐ Cardiac veins



[25/06/2026 11:30 PM] hssnd: Dr. Doc:

☐ VI. Internal Anatomy: The Four Chambers

The heart consists of four chambers:

- ♥ Right Atrium (RA)
- ♥ Right Ventricle (RV)
- ♥ Left Atrium (LA)
- ♥ Left Ventricle (LV)

The atria are receiving chambers, whereas the ventricles are pumping chambers.

♥ 1. Right Atrium (RA)

The right atrium receives deoxygenated blood from:

↑ Superior Vena Cava (SVC)

↓ Inferior Vena Cava (IVC)

□ Coronary Sinus (CS)

□ Sinus Venarum: The smooth posterior part of the right atrium.

Receives: SVC, IVC, Coronary sinus

🧠 Embryological Origin

Derived from the: □ Sinus venosus

👉 Pectinate Muscles

Rough muscular ridges found in:

Anterior wall of right atrium

Function:

Increase the force of contraction.

◆ Crista Terminalis

A muscular ridge extending between:

↑ SVC opening

↓ IVC opening

Function:

Separates the smooth sinus venarum from the rough pectinate region.

🖼️ Sulcus Terminalis

External groove corresponding to the crista terminalis.

☆ Clinical Importance: The superior end marks the location of the ⚡ Sinoatrial (SA) Node, the SA node is the natural pacemaker of the heart.

🕒 Fossa Ovalis: Oval depression located on the interatrial septum.

🧠 Embryological Remnant

Remnant of the: □ Foramen ovale

📖 Valve of Inferior Vena Cava

Also called: ✂ Eustachian Valve

Small fold associated with the opening of the IVC.

📖 Valve of Coronary Sinus

Also called: ✂ Thebesian Valve

Small fold associated with the opening of the coronary sinus.

Right Auricle

Ear-shaped muscular pouch projecting from the atrium.

Functions:

- ☒ Increases atrial capacity
- ☒ Represents the primordial atrium
- ☒ Overlaps the ascending aorta

☐ Venae Cordis Minimae (Thebesian Veins) is Smallest veins of the heart.

Characteristics:

Drain directly into heart chambers.

Most numerous in the atria.

2. Right Ventricle (RV)

The right ventricle forms:

- ☒ Most of the anterior surface
- ☒ Most of the inferior border

Trabeculae Carneae

Irregular muscular ridges on the ventricular wall.

Functions:

Strengthen ventricular contraction.

Prevent suction effects during contraction.

 Conus Arteriosus (Infundibulum), Smooth outflow tract leading to:

- ☐ Pulmonary trunk

Supraventricular Crest

Prominent muscular ridge.

Separates:

-  Rough inflow portion
- from
-  Smooth outflow portion

Papillary Muscles

Three groups:

-  Anterior Papillary Muscle
-  Posterior Papillary Muscle
-  Septal Papillary Muscle

☐ Chordae Tendineae

Strong fibrous cords connecting:

 Papillary muscles

to

Cusps of tricuspid valve

Function:

Prevent prolapse and regurgitation of valve cusps during ventricular systole.

Septomarginal Trabecula

Also called:

 Moderator Band

Muscular bridge extending from:

Interventricular septum

to

Anterior papillary muscle

Function

Carries:

 Right bundle branch of AV bundle

☆ High Yield

Moderator band is unique to the right ventricle.

♡ 3. Left Atrium (LA)

Receives oxygenated blood from:

☐ Four pulmonary veins

(Usually two from each lung)

General Features

☒ Forms most of the base of the heart

☒ Mostly smooth-walled

☒ Slightly thicker wall than right atrium

Contains:

 Pectinate muscles

Functions:

☒ Increases atrial capacity

☒ Represents remnants of primordial atrium

☒ Overlaps the root of the pulmonary trunk

☐ Interatrial Septum

Separates:

♡ Right atrium

from

♡ Left atrium

Contains:

 Semilunar depression corresponding to the floor of the fossa ovalis.

♡ 4. Left Ventricle (LV)

The left ventricle forms:

- ♡ Apex of heart
- ↶ Most of left border
- ↓ Most of diaphragmatic surface

🔗 General Features

The left ventricle is the strongest chamber of the heart.

Reason:

- 🌐 Pumps blood through systemic circulation.

☆ Wall Thickness

Left ventricular wall is:

- 🔧 2–3 times thicker than the right ventricular wall.

Reason:

Higher pressure is required to pump blood throughout the body.

[25/06/2026 11:30 PM] hssnd: 🔗 Trabeculae Carneae

Characteristics:

- ✓ More numerous
- ✓ Finer

than those of the right ventricle.

🔗 Papillary Muscles

Only two papillary muscles:

- 1 Anterior Papillary Muscle
- 2 Posterior Papillary Muscle

Characteristics:

Larger than those of RV.

Attached to mitral valve via chordae tendineae.

📖 Aortic Vestibule

Smooth outflow tract leading to:

- ☐ Aortic orifice
- and
- ☐ Ascending aorta

👤 Ligamentum Arteriosum

Fibrous band connecting:

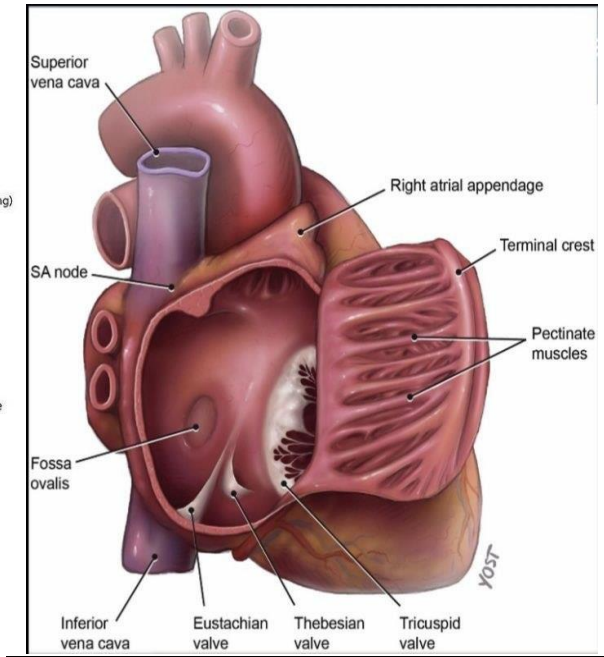
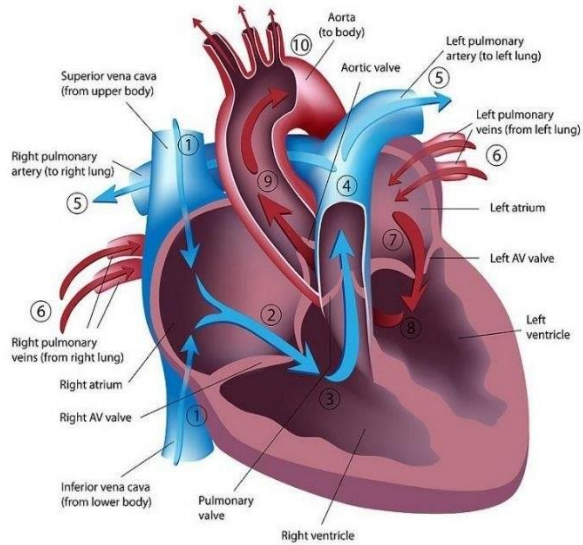
- ☐ Pulmonary trunk
- and
- ☐ Aortic arch

Embryological Remnant

Remnant of:

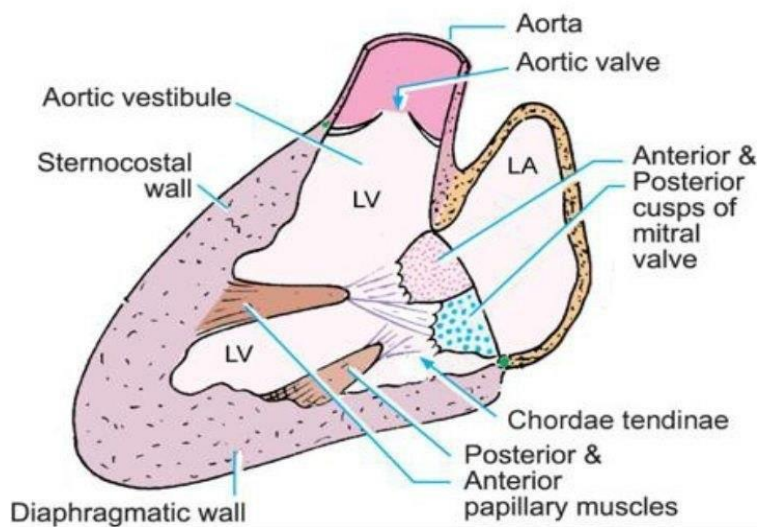
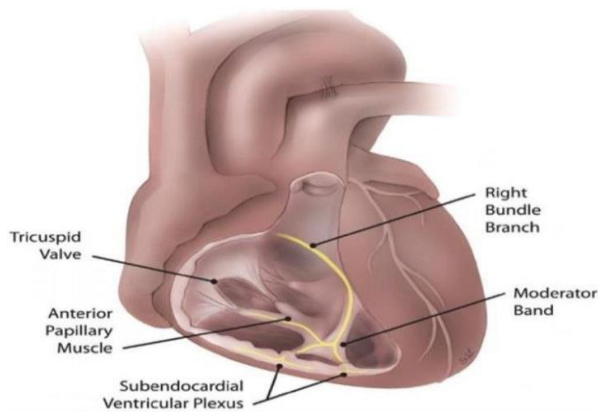
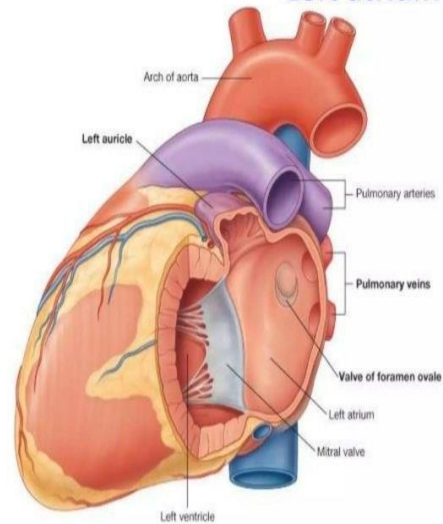
- ☐ Ductus arteriosus

The pathway of blood flow through the heart



Left atrium

A



♥ VII. HEART VALVES AND HEART SOUNDS

Heart valves ensure one-way blood flow through the heart by opening and closing in response to pressure changes.

Atrioventricular (AV) Valves

Located between:

♥ Atria

and

♥ Ventricles

☒ Function:

Prevent backflow of blood from ventricles into atria during ventricular contraction.

Tricuspid Valve

Located between:

♥ Right Atrium

and

♥ Right Ventricle

Structure:

Consists of three cusps:

 1 Anterior cusp

 2 Posterior cusp

 3 Septal cusp

Mitral Valve (Bicuspid Valve)

Located between:

♥ Left Atrium

and

♥ Left Ventricle

Structure:

Consists of two cusps:

 1 Anterior cusp

 2 Posterior cusp

☐ Components of AV Valves

Valve Cusps: Thin fibrous flaps.

☐ Chordae Tendineae: Connect cusps to papillary muscles.

 Papillary Muscles: Prevent valve prolapse during systole.

Semilunar Valves

Located at the origins of the great arteries.

☒ **Function:**
Prevent backflow from arteries into ventricles.

☐ Pulmonary Valve

Located at:

☐ Opening of pulmonary trunk

Cusps:

☐ 1 Anterior cusp

☐ 2 Right cusp

☐ 3 Left cusp

Aortic Valve

Located at:

☐ Opening of ascending aorta

Cusps:

☐ 1 Right cusp

☐ 2 Left cusp

☐ 3 Posterior cusp (Non-coronary cusp)

☆ Features of Semilunar Valves

☒ Pocket-shaped cusps

☒ No chordae tendineae

☒ No papillary muscles

☐ Aortic Sinuses

Three dilatations located behind the cusps of the aortic valve.

☐ 1 Right Aortic Sinus

Gives rise to:

☐ Right Coronary Artery (RCA)

☐ 2 Left Aortic Sinus

Gives rise to:

□ Left Coronary Artery (LCA)

③ Posterior Aortic Sinus

Also called:

✦ Non-coronary sinus

✗ No coronary artery arises from it.

🔊 Heart Sounds

♥ First Heart Sound (S1) — “Lub”

Produced by closure of:

📖 Tricuspid valve

and

📖 Mitral valve

Occurs:

⚡ Beginning of ventricular systole

♥ Second Heart Sound (S2) — “Dup”

Produced by closure of:

📖 Pulmonary valve

and

📖 Aortic valve

Occurs:

⚡ End of ventricular systole

□ VIII. CORONARY CIRCULATION

The heart is supplied by two major coronary arteries □:

① Right Coronary Artery (RCA)

② Left Coronary Artery (LCA)

Both arise from the respective:

→ Right and left aortic sinuses of the ascending aorta □ ↑

● RIGHT CORONARY ARTERY (RCA)

The RCA arises from the right aortic sinus and runs in the coronary groove around the heart .

Its branches include:

1 ⚡ Sinu-atrial (SA) Nodal Branch

- Usually arises near the origin of the RCA.
 - Supplies the SA node (natural pacemaker of the heart) ⚡ .
 - Supplies the SA node in approximately 60% of people.
-

2 Right Marginal Branch

- Supplies the lateral part of the right side of the heart .
 - Mainly supplies the right ventricle.
-

3 ⚡ Atrioventricular (AV) Nodal Branch

- Arises at the “crux” of the heart (junction of septa).
 - Supplies the AV node ⚡.
 - Supplies the AV node in about 80% of people.
-

4 Posterior Interventricular Branch

- Also called the Posterior Descending Artery (PDA).
 - Runs in the posterior interventricular groove toward the apex .
 - Supplies areas of both ventricles.
 - Gives off interventricular septal branches.
-

● LEFT CORONARY ARTERY (LCA)

The LCA arises from the left aortic sinus and passes between the left auricle and pulmonary trunk before dividing into its terminal branches .

1 Anterior Interventricular Branch

(Also called LAD — Left Anterior Descending artery)

- Passes along the anterior interventricular groove toward the apex.
 - Supplies a large part of the anterior heart surface and left ventricle.
-

2 Lateral (Diagonal) Branch

- Usually arises from the anterior interventricular branch (LAD).
 - Supplies ventricular walls .
-

3 Septal Branches

- Supply the anterior two-thirds of the interventricular septum.
- Also supply parts of the conduction system .

[25/06/2026 11:31 PM] hssnd:

4 Circumflex Branch

- Follows the coronary groove around the left side of the heart.
 - Continues toward the posterior surface of the heart.
-

5 Left Marginal Artery

- Arises from the circumflex branch.
 - Supplies the left ventricle .
-

6 SA Nodal Branch

- In about 40% of individuals, the artery supplying the SA node arises from the circumflex branch instead of the RCA.
-

IMPORTANT NOTE ☐

Coronary arteries supply oxygen-rich blood to the myocardium  ☐.
Blockage of these arteries can reduce blood flow and may cause:

 Ischemia

 Myocardial infarction (heart attack)  ☐

☆ Coronary Dominance

Determined by: Origin of Posterior Descending Artery (PDA)

Right Dominance

PDA arises from RCA.

Occurs in:

 ~67%

Left Dominance

PDA arises from LCA.

Occurs in:

 ~15%

Codominance

Both arteries contribute.

Occurs in:

 ~18%

☐ IX. Venous Drainage of the Heart

Most venous blood drains into the: Coronary Sinus

Location:

Posterior coronary sulcus

Opens Into  Right atrium

Between:

 IVC opening

and

 Tricuspid valve

Tributaries of Coronary Sinus

☐ Great Cardiac Vein

Accompanies:

☐ LAD

Drains:

 Anterior heart

☐ Middle Cardiac Vein

Accompanies:

☐ PDA

Drains:

♡ Posterior heart

☐ Small Cardiac Vein

Accompanies:

☐ Right marginal artery

Drains:

♡ Right side of heart

☐ Oblique Vein of Left Atrium

Runs on posterior wall of left atrium.

Joins:

☐ Great cardiac vein

to form:

☐ Coronary sinus

☐ Anterior Cardiac Veins

Drain directly into:

♡ Right atrium

✗ Do not enter coronary sinus.

☐ Venae Cordis Minimae (Thebesian Veins)

Smallest veins of heart.

Drain directly into:

♡ All four chambers

Most numerous in atria.

X. Innervation of the Heart

The innervation of the heart is primarily managed by the Autonomic Nervous System (ANS)

☐  ☐, which regulates:

♡ Heart rate

 Force of contraction

☐ Overall cardiac output

Although the heart can beat without nervous system stimulation due to its own conducting system ⚡, the ANS acts as the “accelerator” and “brake” 🚗 to adjust heart activity according to the body’s needs.

THE CARDIAC PLEXUS

The autonomic fibers reach the heart through a network called the cardiac plexus 🌀☐.

📍 Location:

- Located on the anterior surface of the bifurcation of the trachea.

☐ Composition:

- Sympathetic fibers ⚡
 - Parasympathetic fibers 🌿
 - Visceral afferent (sensory) fibers ☐
-

1⚡ SYMPATHETIC INNERVATION (“The Accelerator”) 🌀

Sympathetic stimulation prepares the body for action by increasing heart activity ❤️☐↑.

💎 Origin:

- Arises from T2–T4 segments of the spinal cord ☐.

💎 Distribution:

Fibers pass through:

- Middle cervical ganglia
- Cervico-thoracic (stellate) ganglia
- First four thoracic sympathetic chain ganglia

Then they pass through the cardiac plexus 🌀 to reach:

- SA node ⚡
- Cardiac muscle (myocardium) ❤️ of atria and ventricles

🌀 Effects:

❤️ SA Node:

- Increases the rate of contraction → ↑ heart rate

🫀 Cardiac Muscle:

- Increases force of contraction
- Increases ventricular pressure
- Increases stroke volume ☐

⚠️ Sympathetic nerves are especially dominant in ventricular muscle, increasing contraction strength.

☐ Coronary Arteries:

- Causes dilation of coronary arteries
- Increases blood flow to active heart muscle ❤️

2 PARASYMPATHETIC INNERVATION (“The Brake”)

Parasympathetic stimulation dominates during rest and conserves energy .

◆ Origin:

- Reaches the heart through branches of the Vagus nerve (CN X) □.

◆ Distribution:

Main targets:

- SA node ⚡
- AV node ⚡
- Atrial muscle ♥

 Effects:

♥ Heart Rate:

- Decreases SA node activity
- Slows heart rate

⚡ Conduction:

- Decreases excitability of AV node
- Slows electrical transmission to ventricles

 Force of Contraction:

- Reduces force of myocardial contraction

□ Target Dominance:

- Mainly controls conducting tissues (SA and AV nodes).

[25/06/2026 11:31 PM] hssnd:  Clinical Note:

Strong vagal stimulation at the AV node may cause AV block ⚡ ⊗.

□ Coronary Arteries:

- Causes constriction of coronary arteries.

3 SENSORY INNERVATION AND PAIN

The heart and its coverings have sensory fibers that carry information back to the CNS □.

♥ Fibrous & Parietal Pericardium:

- Supplied by phrenic nerves (C3–C5).

 Referred Pain:

Because they share spinal segments with the shoulder skin, pericardial pain may be felt in the ipsilateral supraclavicular region (top of the shoulder on the same side) 🤔 ➡️ □.

♡ Visceral Pericardium (Epicardium):

- Supplied by the same autonomic nerves as the heart.
- Generally not sensitive to pain.

👂 XI. Surface Anatomy and Auscultation

🎵 Aortic Valve Area

📍 Right 2nd intercostal space

📍 Right sternal border

🎵 Pulmonary Valve Area

📍 Left 2nd intercostal space

📍 Left sternal border

🎵 Tricuspid Valve Area

📍 Left lower sternal border

📍 4th–5th intercostal space

🎵 Mitral Valve Area

📍 Apex of heart

📍 Left 5th intercostal space

📍 Midclavicular line

🧠 XII. Important Embryological Remnants

Adult Structure → Embryological Remnant

● Fossa Ovalis → Foramen Ovale

🌀 Ligamentum Arteriosum → Ductus Arteriosus

□ Sinus Venarum → Sinus Venosus

👂 Right Auricle → Primordial Atrium

👂 Left Auricle → Primordial Atrium

★ XIII. HIGH-YIELD EXAM PEARLS

- ☆ Left ventricle wall is 2–3 times thicker than right ventricle wall.
- ☆ Right ventricle forms most of the anterior surface of the heart.
- ☆ Left ventricle forms the apex.
- ☆ Left atrium forms most of the base.
- ☆ Coronary sinus drains most venous blood from the heart.
- ☆ Moderator band carries the right bundle branch.
- ☆ Semilunar valves have no chordae tendineae.
- ☆ Semilunar valves have no papillary muscles.
- ☆ Coronary dominance depends on the origin of the PDA.
- ☆ Great cardiac vein accompanies LAD.
- ☆ Middle cardiac vein accompanies PDA.
- ☆ Small cardiac vein accompanies right marginal artery.
- ☆ SA node lies near the superior end of the sulcus terminalis.
- ☆ Pericardial pain is carried by the phrenic nerve (C3–C5) and is referred to the shoulder.
- ☆ Ligamentum arteriosum is the remnant of the ductus arteriosus.
- ☆ Fossa ovalis is the remnant of the foramen ovale.
- ☆ RCA supplies the SA node in about 60% of individuals.
- ☆ RCA supplies the AV node in about 80% of individuals.
- ☆ LAD supplies the anterior two-thirds of the interventricular septum.
- ☆ The transverse pericardial sinus is an important surgical landmark used to control blood flow through the aorta and pulmonary trunk during cardiac surgery.

BLOOD VESSELS AND LYMPHATIC SYSTEM NOTE

♡ BLOOD VESSELS & LYMPHATIC SYSTEMS

□ 1. Core Circulatory Principles

The heart functions as a double pump □ that maintains blood flow through two major circuits:

A. Pulmonary Circulation

Pulmonary circulation transports oxygen-poor blood from the heart to the lungs for gas exchange.

□ Blood pathway:

Right ventricle → Pulmonary trunk → Right & left pulmonary arteries → Pulmonary capillaries → Pulmonary veins → Left atrium

◆ The right ventricle pumps deoxygenated blood into the pulmonary trunk.

◆ The pulmonary trunk divides into:

- Right pulmonary artery
- Left pulmonary artery

◆ In the lung capillaries:

- Carbon dioxide (CO₂) is removed from blood
- Oxygen (O₂) enters the blood

◆ Oxygen-rich blood returns to the left atrium through the four pulmonary veins.

✧ Important: Pulmonary arteries are unique because they carry deoxygenated blood, while pulmonary veins carry oxygenated blood.

♡ B. Systemic Circulation

Systemic circulation delivers oxygen-rich blood from the heart to the entire body.

[25/06/2026 11:31 PM] hssnd: □ Blood pathway:

Left ventricle → Aorta → Body tissues → Venae cavae → Right atrium

◆ Oxygenated blood leaves the left ventricle through the aorta, the largest artery in the body.

◆ Arteries distribute blood to tissues where:

- Oxygen and nutrients are delivered
- Carbon dioxide and wastes are collected

◆ Venous blood returns to the heart through:

- Superior vena cava → drains head, neck, upper limbs, and thorax
- Inferior vena cava → drains abdomen, pelvis, and lower limbs
- Coronary sinus → drains the heart muscle

◆ These vessels empty into the right atrium.

□ 2. The Systemic Arterial System

The systemic arteries arise from the aorta and supply different regions of the body.

☐ A. Head and Neck

Four main arterial pairs supply this region:

● Common Carotid Arteries

- Right common carotid → arises from the brachiocephalic trunk
- Left common carotid → arises directly from the aortic arch

At the neck, they divide into:

◆ External Carotid Artery

Supplies:

- Face
- Scalp
- Neck structures

◆ Internal Carotid Artery

Supplies:

- Brain
- Eye structures

Together with the vertebral arteries, it provides the main blood supply to the brain ☐.

● Subclavian Artery Branches

The subclavian artery gives important branches:

- Vertebral artery → supplies brain and spinal cord
- Thyrocervical trunk → supplies neck and thyroid region
- Costocervical trunk → supplies deep neck and upper thorax

☐ B. The Thorax

The thoracic wall receives arterial blood from three main sources:

● 1. Thoracic Aorta

The thoracic aorta gives branches including:

- Posterior intercostal arteries → supply spaces between ribs
- Subcostal arteries → supply the lower thoracic and abdominal wall

● 2. Subclavian Artery

The subclavian artery gives rise to:

☐ Internal Thoracic Artery

(formerly called internal mammary artery)

Supplies:

- Anterior chest wall
- Ribs
- Skin
- Mammary glands

● 3. Axillary Artery

Provides:

- Superior thoracic artery

- Lateral thoracic artery

These supply the lateral and upper thoracic regions.

C. Upper Limbs

The arterial supply of the upper limb follows a continuous pathway with different names depending on location:

→ Subclavian artery → Axillary artery → Brachial artery → Radial & Ulnar arteries

◆ Subclavian Artery

Becomes the axillary artery when it enters the armpit.

◆ Axillary Artery

Supplies:

- Shoulder region
- Upper chest
- Upper limb

◆ Brachial Artery

Main artery of the arm.

At the elbow, it divides into:

□ Radial artery

- Supplies lateral forearm
- Contributes to hand circulation

□ Ulnar artery

- Supplies medial forearm
- Contributes to hand circulation

These arteries form:

👉 Palmar arches → supply the hand and fingers.

□ D. The Abdomen

The abdominal aorta extends from the diaphragm to approximately L4 vertebral level.

It gives:

◆ Unpaired (Midline) Branches

1. Celiac Trunk

Supplies the foregut:

Includes:

- Stomach
- Liver
- Gallbladder
- Pancreas
- Spleen
- Duodenum

It divides into three major branches:

◆ Left gastric artery

◆ Splenic artery

◆ Common hepatic artery

2. Superior Mesenteric Artery (SMA)

Supplies the midgut:

Includes:

- Most of small intestine
- Cecum
- Appendix
- Ascending colon
- Part of transverse colon

3. Inferior Mesenteric Artery (IMA)

Supplies the hindgut:

Includes:

- Distal colon
- Sigmoid colon
- Rectum

Paired Branches

☐ Renal arteries

Supply the kidneys.

☐ Suprarenal arteries

Supply adrenal glands.

♂ Gonadal arteries

Supply:

- Testes in males
- Ovaries in females
- ☐ Lumbar arteries

Four pairs supplying the abdominal wall.

Median Sacral Artery

A small unpaired artery arising from the lower aorta.

Supplies:

- Sacrum
- Coccyx

☐ E. Pelvis and Lower Limbs

At approximately L4, the abdominal aorta divides into:

☐ Right and left common iliac arteries

[25/06/2026 11:31 PM] hssnd: These divide into:

Internal Iliac Artery

Supplies:

- Pelvic organs
- Pelvic walls
- Buttocks
- Medial thigh

External Iliac Artery

Continues into the lower limb and becomes:

☐ Femoral artery

The femoral artery becomes:

☐ Popliteal artery (behind the knee)

Then divides into:

- ☐ Anterior tibial artery
→ becomes dorsalis pedis artery at ankle
 - ☐ Posterior tibial artery
→ supplies posterior leg and foot
-

3. Specialized Concept: Anastomoses

An anastomosis is a connection between branches of arteries supplying the same area.

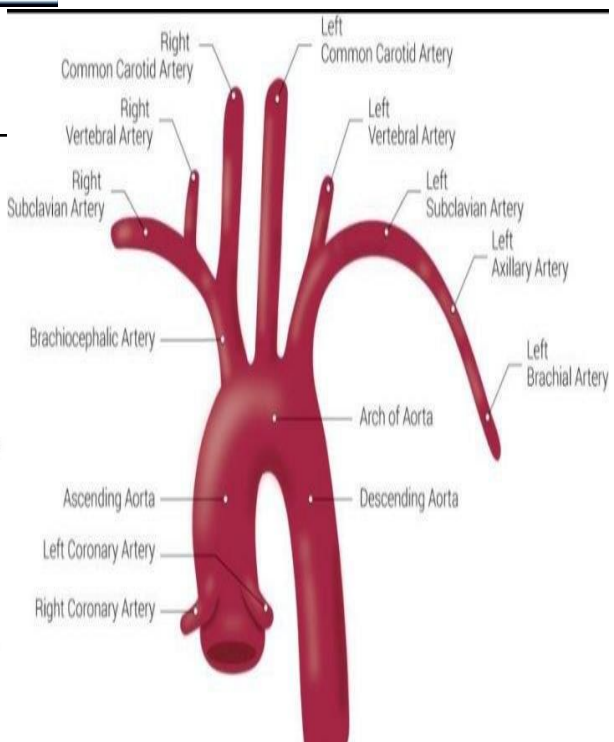
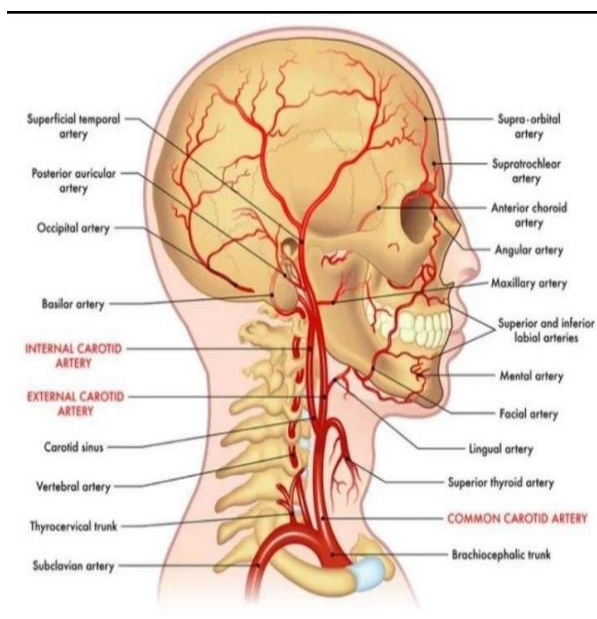
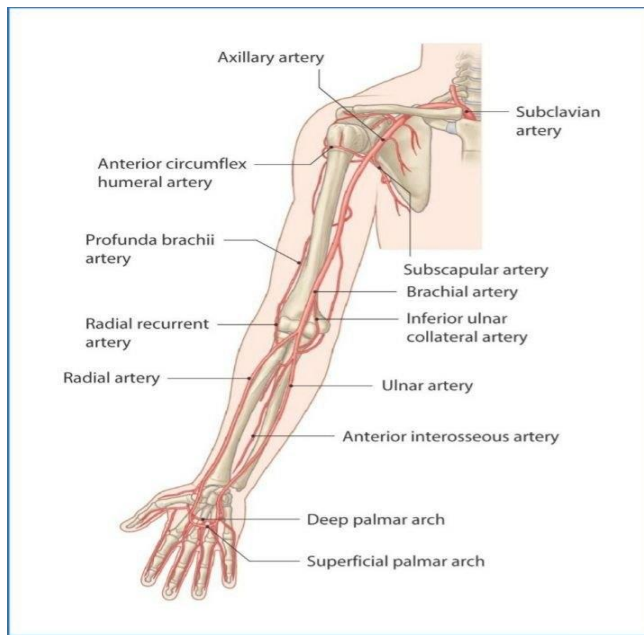
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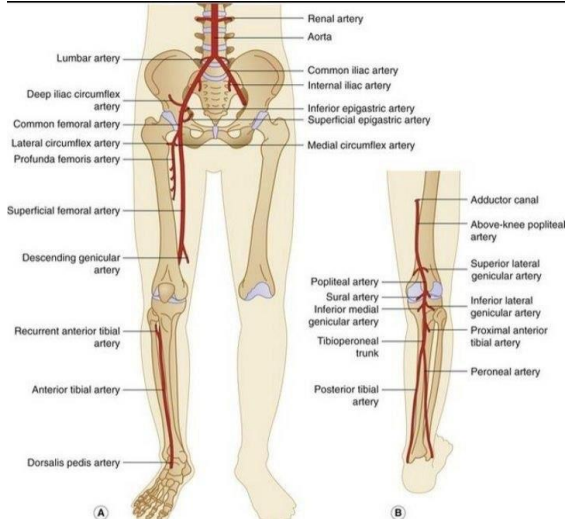
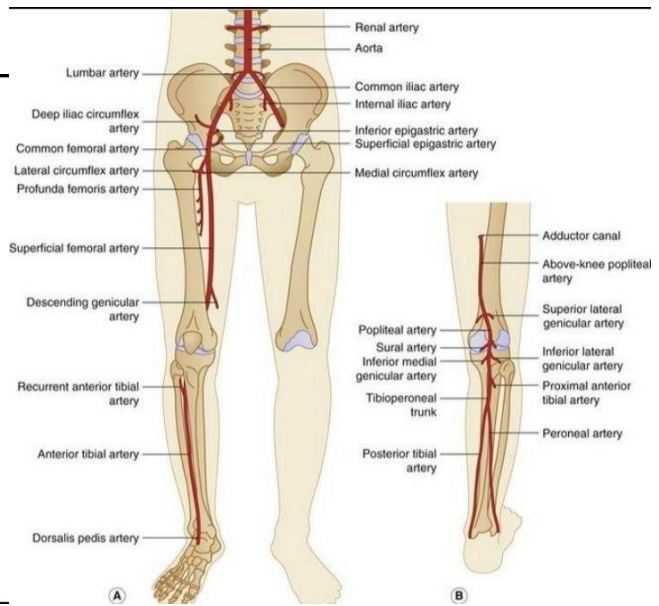
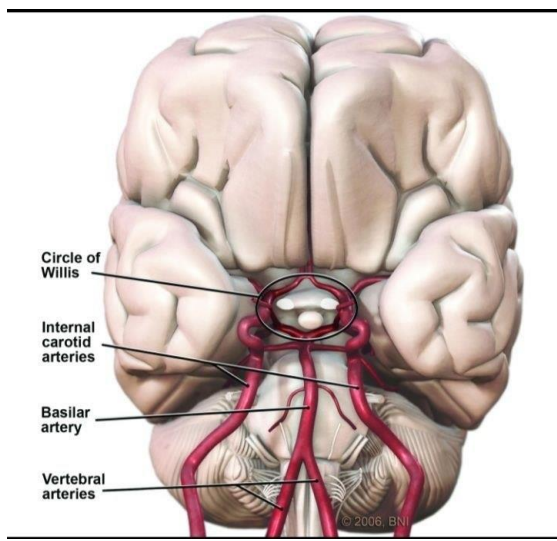
- ☒ Provides alternative blood routes
- ☒ Maintains blood supply if one vessel is blocked

This alternative pathway is called:

- ☐ Collateral circulation

☐ Arteries without significant connections are called:
End arteries





THORACIC CAVITY – ARTERIES

The arteries of the thoracic cavity arise mainly from the thoracic (descending) aorta and its branches. They supply the thoracic wall, viscera (heart, lungs, esophagus), and associated structures.

1 THORACIC AORTA

- Continuation of the arch of the aorta.
- Enters the thorax through the aortic hiatus of the diaphragm at the level of T12.
- Descends in the posterior mediastinum in front of the vertebral bodies.
- Ends at the level of T12 by passing through the diaphragm and continues as the abdominal aorta.

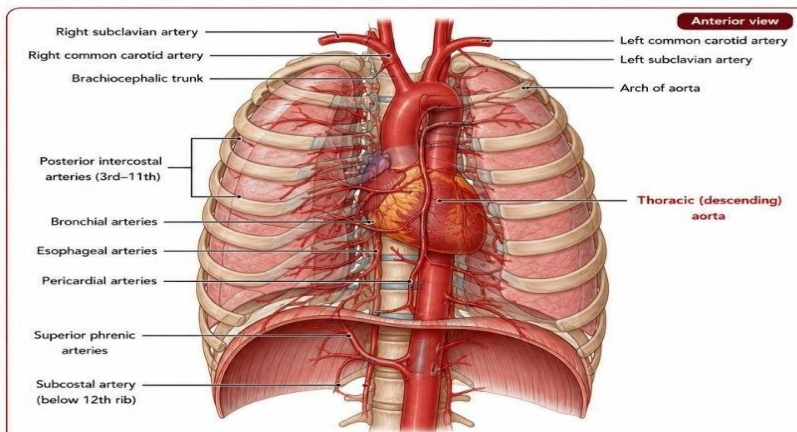
2 BRANCHES OF THE THORACIC AORTA

A Parietal (to thoracic wall)

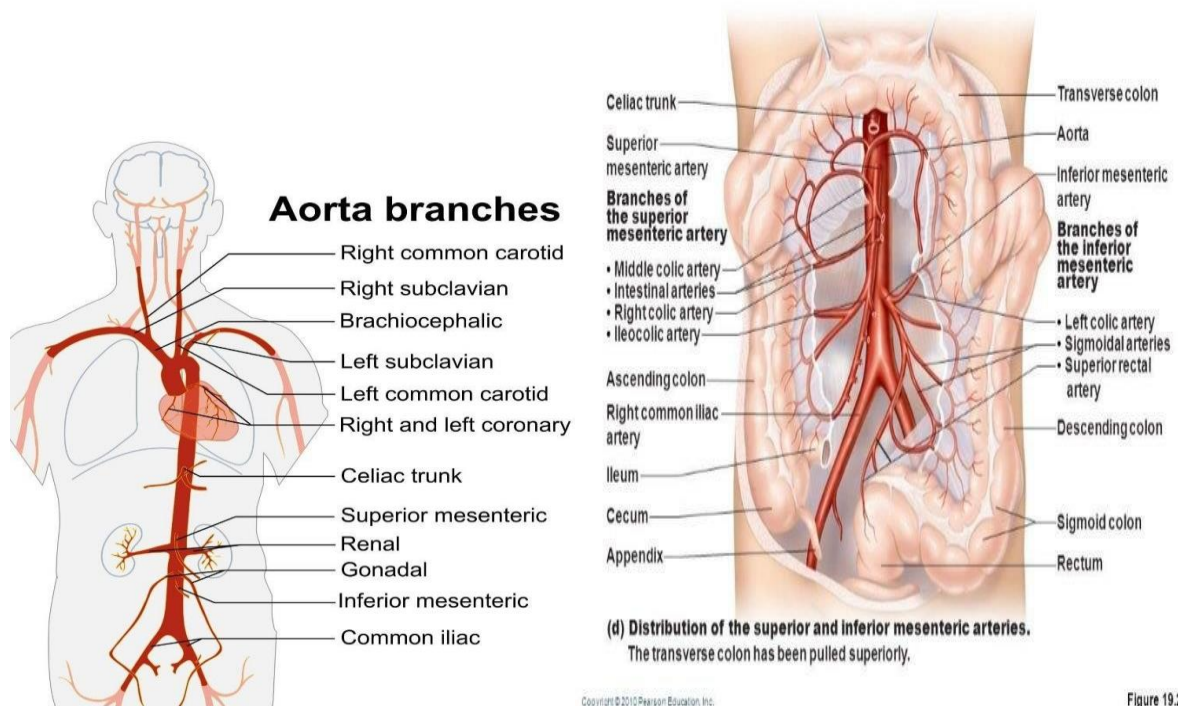
- Posterior intercostal arteries (3rd–11th)
- Subcostal artery (below 12th rib)
- Superior phrenic arteries

B Visceral (to thoracic organs)

- Bronchial arteries
- Esophageal arteries
- Pericardial arteries
- Mediastinal arteries



KEY POINT: The thoracic aorta gives parietal branches to the thoracic wall and visceral branches to the thoracic organs.



[25/06/2026 11:33 PM] hssnd: Dr. Doc:

□ 4. Venous Drainage

Veins are blood vessels that return blood back to the heart ♡.
They are divided into:

◆ Superficial veins

- Located just beneath the skin
- Can often be seen easily
- Important clinically for:

Blood collection 🩺
Injections

◆ Deep veins

- Located deeper in the body
- Usually travel with arteries
- Often have the same names as arteries

☐ Important: Veins contain valves that prevent backward movement of blood, especially in the lower limbs where blood must move against gravity.

☐ A. Veins of Head and Neck

The head and neck are mainly drained by:

◆ External Jugular Vein

Drains:

- Scalp
- Parts of face
- Superficial neck

☐ Empties into the subclavian vein

◆ Internal Jugular Vein

Drains:

- Brain
- Meninges
- Deep face
- Deep neck structures

It is:

- ☒ Larger
- ☒ Deeper

than the external jugular vein.

The internal jugular vein joins the subclavian vein to form:

☐ Brachiocephalic vein

The two brachiocephalic veins join to form:

☐ Superior vena cava → Right atrium

☐ Dural Sinuses

Most blood from the brain drains into dural sinuses.

They are special channels located between layers of the dura mater.

Main sinuses include:

- Superior sagittal sinus
- Inferior sagittal sinus
- Straight sinus
- Transverse sinus
- Sigmoid sinus

The sigmoid sinus continues as:

☐ Internal jugular vein

B. Veins of Upper Extremity

The upper limb has:

- Superficial drainage
- Deep drainage

◆ Deep Veins

Deep veins accompany arteries.

Examples:

- ☐ Radial veins
- ☐ Ulnar veins

They drain blood from:

- Deep palmar arch
- Superficial palmar arch

The radial and ulnar veins join to form:

→ Brachial vein

Pathway:

Brachial → Axillary → Subclavian → Brachiocephalic → Superior vena cava

◆ Superficial Veins

Main superficial veins:

♥ Cephalic Vein

Drains:

- Lateral side of hand
- Lateral forearm

→ Joins axillary vein near shoulder

♥ Basilic Vein

Drains:

- Medial side of forearm
- Medial side of arm

→ Joins brachial vein

✍ Median Cubital Vein

Connects:

Cephalic vein ↔ Basilic vein

✂ Clinical importance:

It is a common site for venipuncture (blood sampling).

□ C. Veins of Thorax

The thorax is drained mainly by the:

◆ Azygos System

Includes:

- Azygos vein
- Hemiazygos vein
- Accessory hemiazygos vein

These eventually drain into:

→ Superior vena cava

□ Azygos Vein

Located on the right side of vertebral column.

Receives:

- Right posterior intercostal veins

□ Hemiazygos Vein

Located on the left side.

Receives:

- Left lower posterior intercostal veins

Then crosses to join azygos vein.

□ Accessory Hemiazygos Vein

Drains:

- Upper left thoracic wall

Joins the azygos vein.

📌 Clinical importance:

The azygos system can act as a backup pathway if the inferior vena cava is blocked.

□ D. Veins of Abdomen

Venous blood from the abdomen returns to the heart mainly through the:

➡ Inferior vena cava (IVC)

The IVC drains blood from:

- Abdominal wall
- Pelvis
- Lower limbs
- Abdominal organs

However, blood from some digestive organs does not directly enter the IVC.

Instead, it passes through a special circulation:

🔍 Hepatic Portal System

The hepatic portal system carries blood from digestive organs to the liver before entering general circulation.

It drains blood from:

- Stomach
- Intestines
- Spleen
- Pancreas
- Gallbladder

Main pathway:

➡ Superior mesenteric vein + Splenic vein → Hepatic portal vein → Liver → Hepatic veins
→ Inferior vena cava

📌 Function:

The liver processes absorbed nutrients and removes harmful substances before blood enters the systemic circulation.

□ E. Veins of Lower Extremity

The lower limb has:

- Deep veins
- Superficial veins

◆ Deep Veins

Deep veins accompany arteries.

Main pathway:

Foot → Tibial veins → Popliteal vein → Femoral vein → External iliac vein → Common iliac vein → Inferior vena cava

[25/06/2026 11:33 PM] hssnd: □ Tibial Veins

Includes:

- Anterior tibial veins
- Posterior tibial veins

They drain blood from the foot and leg.

They join to form:

→ Popliteal vein

□ Popliteal Vein

Located behind the knee.

Continues as:

→ Femoral vein

□ Femoral Vein

Receives blood from:

- Deep femoral vein
- Great saphenous vein

Becomes:

→ External iliac vein

◆ Superficial Veins

♥ Great Saphenous Vein

☆ Longest vein in the body

Drains:

- Medial side of foot
- Medial leg
- Medial thigh

Ends in:

→ Femoral vein

♥ Small Saphenous Vein

Drains:

- Lateral side of foot
- Posterior leg

Ends in:

→ Popliteal vein

🧠 5. Fetal Circulation

Fetal circulation is the blood circulation present before birth.

The fetus depends on the placenta for:

👉 Oxygen supply

👉 Nutrients

👉 Waste removal

The fetal lungs, kidneys, and digestive system are not fully functioning before birth.

👉 Placental Exchange

Maternal and fetal blood usually do not directly mix.

Exchange occurs through diffusion across the placenta.

● Umbilical Arteries (2)

Carry:

- Oxygen-poor blood
- Waste products

From fetus → placenta

● Umbilical Vein (1)

Carries:

- Oxygen-rich blood
- Nutrients

From placenta → fetus

♡ Special Fetal Structures

Because the fetal lungs are not functioning, blood bypasses the lungs using special structures.

◆ Foramen Ovale

An opening between:

→ Right atrium → Left atrium

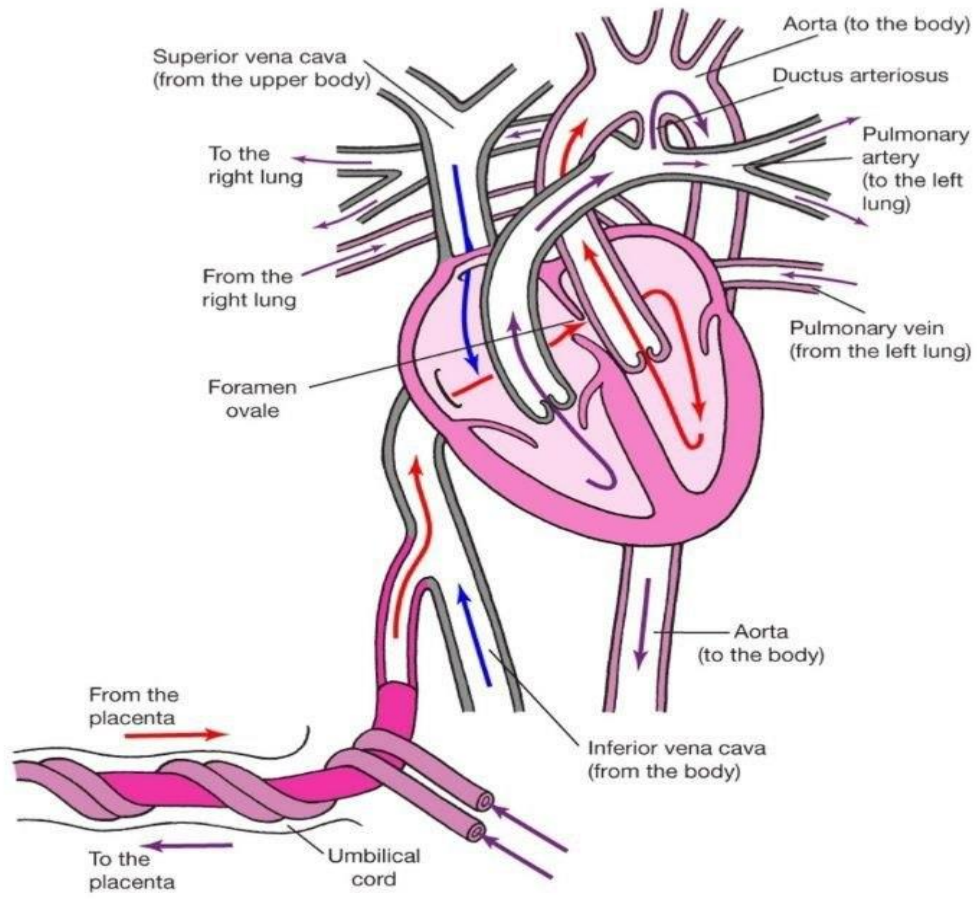
Allows blood to bypass pulmonary circulation.

◆ Fetal Circulation Differences

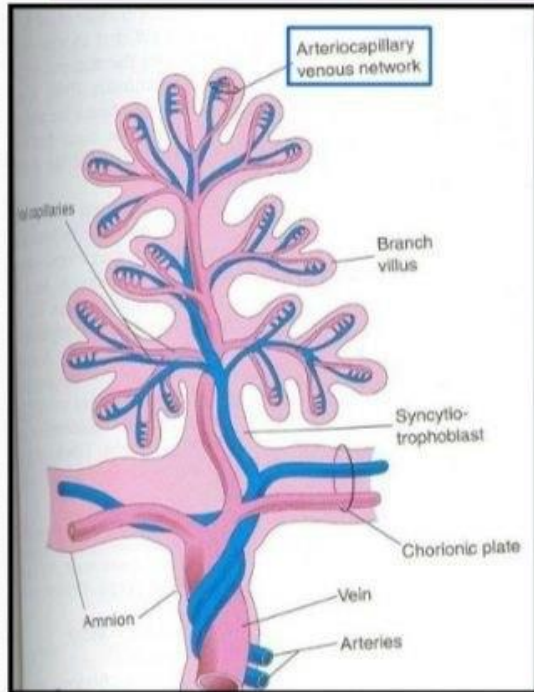
Two major differences:

1 Fetus receives oxygen from placenta, not lungs.

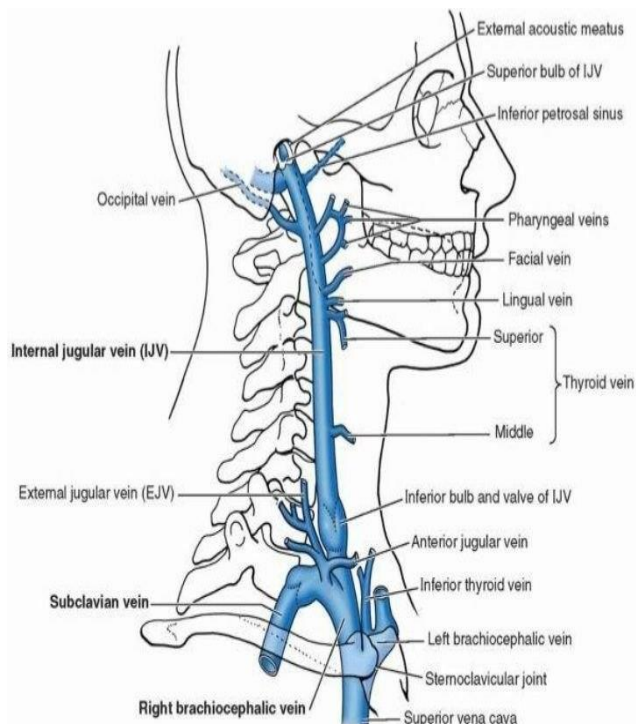
2 Only a small amount of blood goes to the lungs because they are not yet active.



Fetal Placental Circulation



- **Two Umbilical Arteries:**
- Carry poorly oxygenated blood from the fetus to the placenta.
- within the branch chorionic villi, they form:
- **Arterio-Capillary Venous network:**
- It brings the fetal blood extremely close to the maternal blood.
- The well oxygenated fetal blood in the capillaries passes into veins accompanying the chorionic arteries.
- At the umbilical cord, they form the **One Umbilical Vein.**



Major Veins of the Body

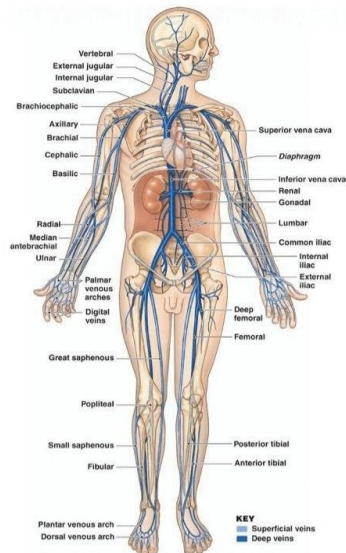
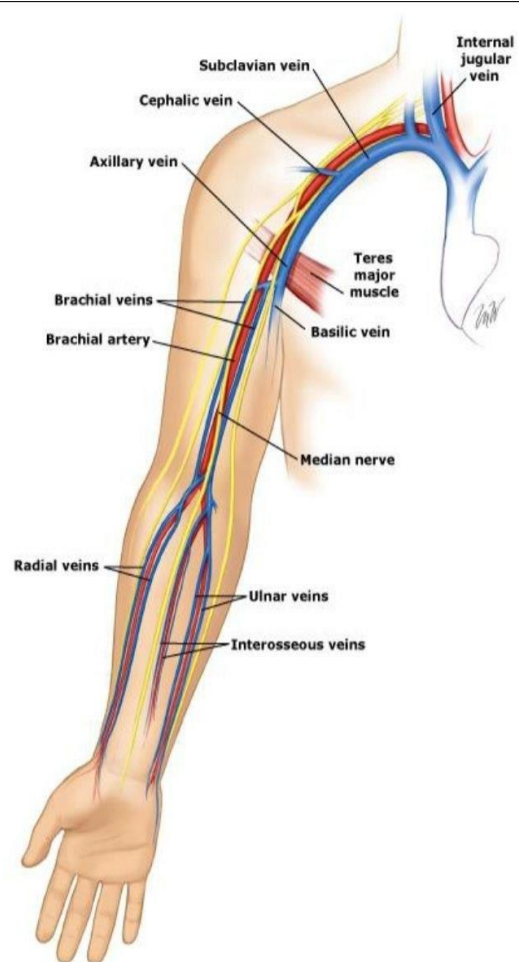
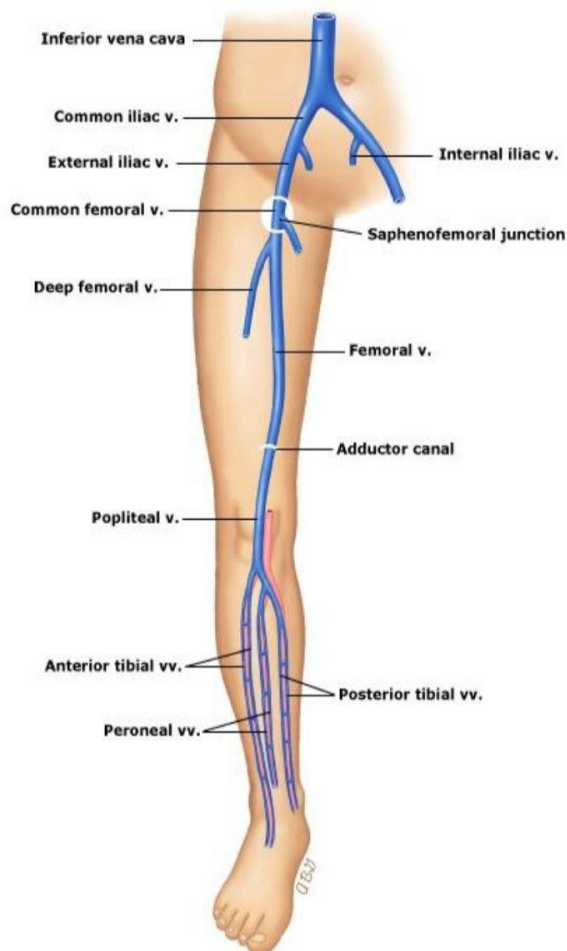
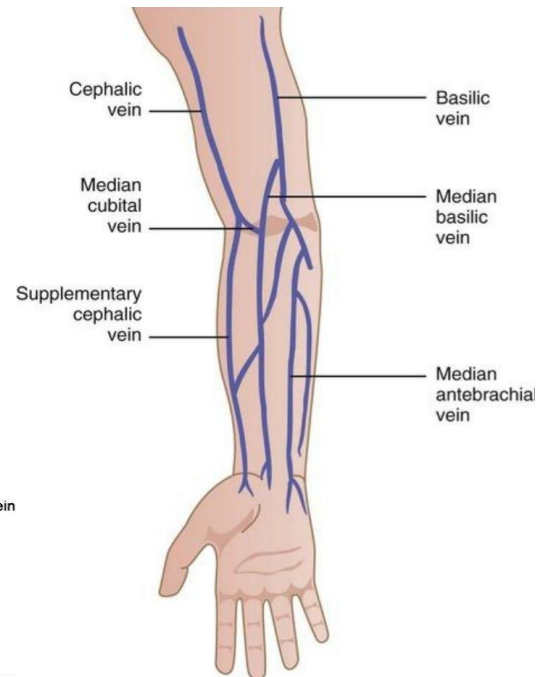
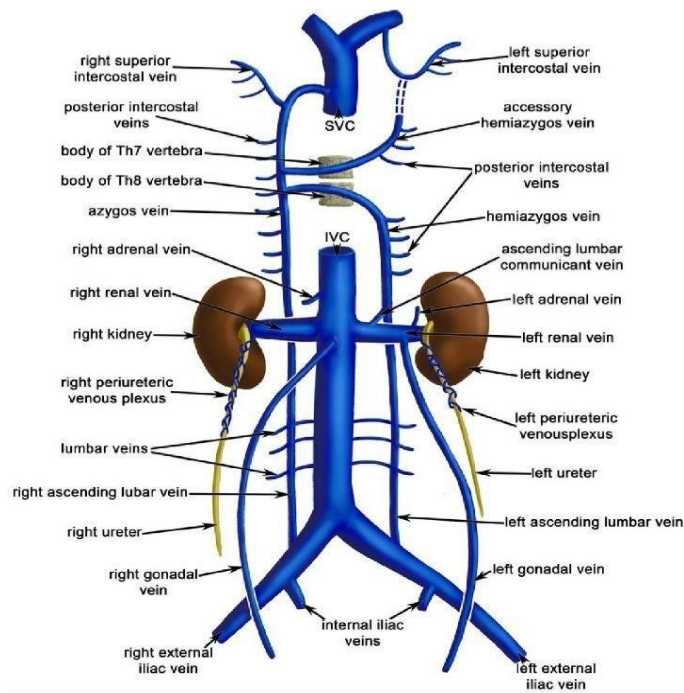
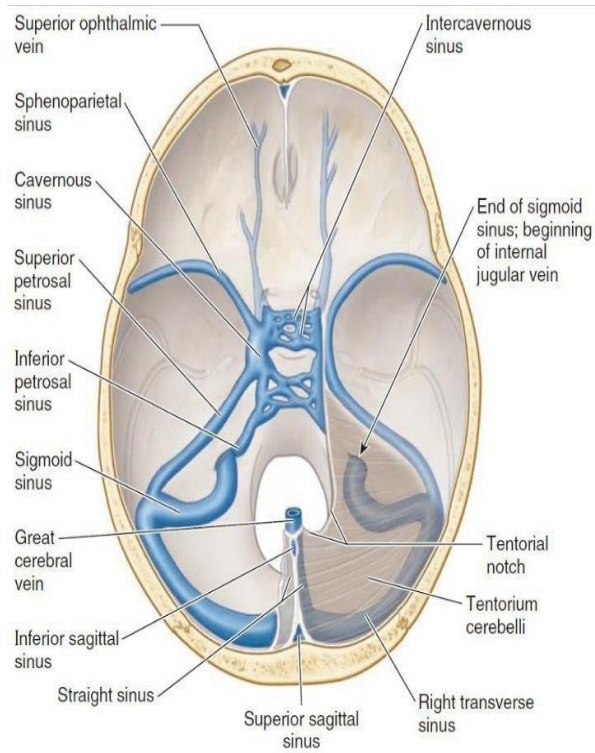


image via: anatomylibrary99.com



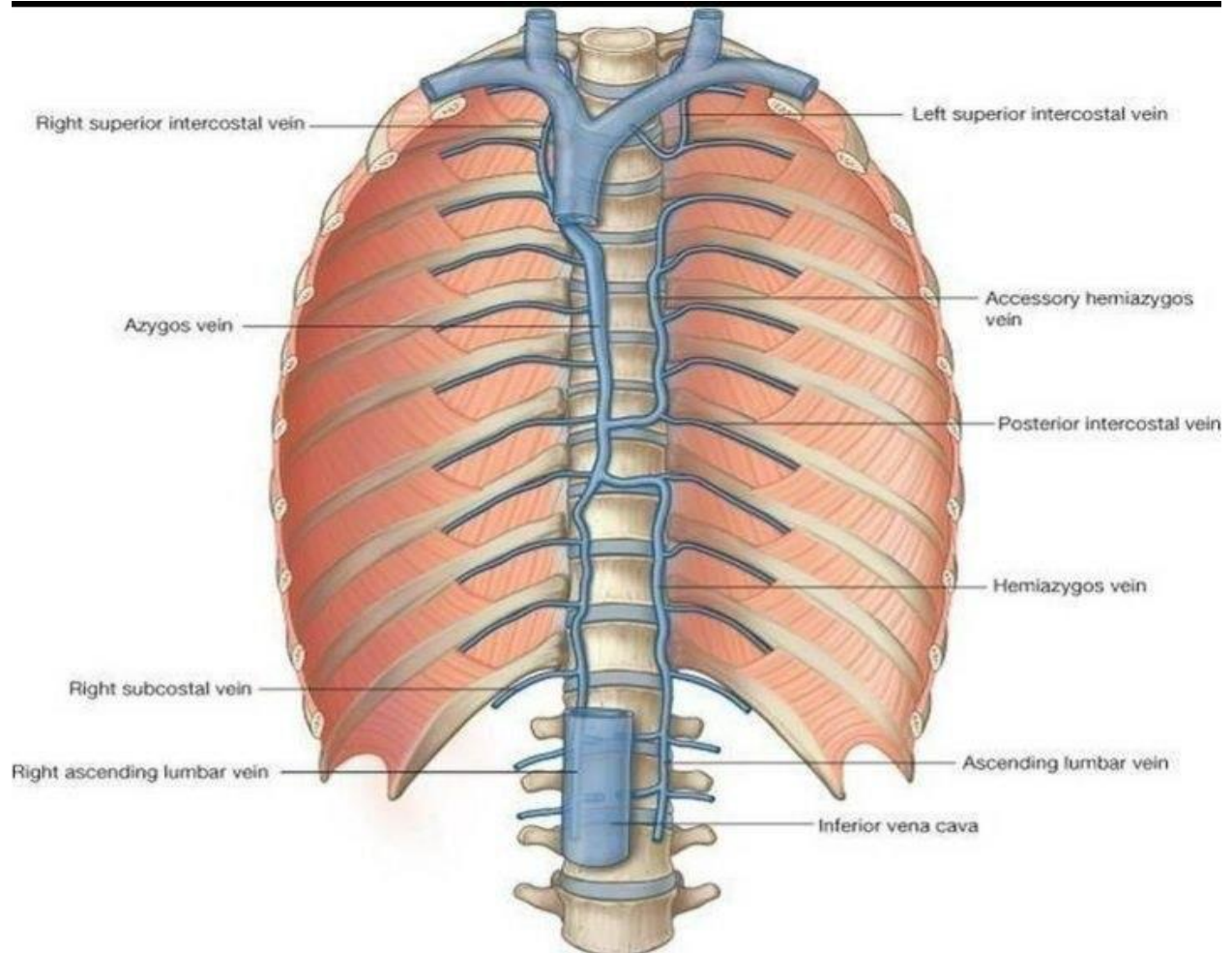


Deep venous drainage of lower limb

- Located in and drains the compartments of the lower limb
- Runs with the corresponding arteries (same name and usually paired)

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6. Pulmonary Circulation

The pulmonary circulation is the pathway that carries blood between the heart and lungs □□.
Its main function is:

☒ Remove carbon dioxide (CO₂)

☒ Add oxygen (O₂) to blood

□ Blood Pathway

Right ventricle → Pulmonary trunk → Pulmonary arteries → Lung capillaries → Pulmonary veins → Left atrium

◆ Blood leaves the right ventricle and enters the pulmonary trunk.

◆ The pulmonary trunk divides into:

- Right pulmonary artery
- Left pulmonary artery

◆ These arteries carry oxygen-poor blood to the lungs.

◆ In the lung capillaries:

- CO₂ leaves the blood and enters alveoli
- O₂ enters the blood from alveoli

◆ Oxygen-rich blood returns through the:

→ Four pulmonary veins

to the:

→ Left atrium

✧ Important: Pulmonary arteries carry deoxygenated blood, while pulmonary veins carry oxygenated blood.

7. Systemic Circulation

Systemic circulation supplies blood to the entire body except the lungs.
Its function:

☒ Deliver oxygen and nutrients

☒ Remove carbon dioxide and wastes

□ Blood Pathway

Left ventricle → Aorta → Arteries → Capillaries → Veins → Right atrium

◆ Oxygen-rich blood leaves the:

→ Left ventricle

through the:

→ Aorta

◆ The aorta branches into smaller arteries that supply:

- Head and neck
- Upper limbs
- Thorax
- Abdomen
- Pelvis
- Lower limbs

◆ At tissue level:

O₂ and nutrients move into cells.

CO₂ and wastes enter blood.

◆ Blood returns to the heart through:

☐ Superior vena cava

Drains:

- Head
- Neck
- Upper limbs
- Thorax

☐ Inferior vena cava

Drains:

- Abdomen
- Pelvis
- Lower limbs

☐ Coronary sinus

Drains:

- Heart muscle

All empty into:

☐ Right atrium

8. Lymphatic System

The lymphatic system is a one-way drainage system that returns excess tissue fluid back to the bloodstream.

It is closely related to the cardiovascular system ☐.

☆ Functions of Lymphatic System

1. Drainage of Tissue Fluid

Blood capillaries lose some fluid into tissues.

The lymphatic system:

☐ Collects excess fluid

☐ Returns it to the blood

This prevents swelling (edema).

2. Absorption of Lipids

Special lymphatic vessels in the small intestine called:

→ Lacteals
absorb dietary fats.

🛡️ 3. Immune Defense

The lymphatic system helps protect the body by:

- Filtering pathogens
- Transporting lymphocytes
- Supporting immune responses

□ Components of Lymphatic System

Includes:

◆ Lymphatic Plexuses

Networks of lymphatic capillaries that begin in tissues.

◆ Lymphatic Vessels

Thin-walled vessels containing many valves.

Functions:

- Carry lymph
- Prevent backward flow

◆ Lymph Nodes

Small structures along lymphatic vessels.

Functions:

- Filter lymph
- Remove foreign materials
- Contain immune cells

◆ Lymphocytes

Immune cells responsible for defense.

Examples:

- B cells
- T cells

🌀 Lymphatic Vessels, Trunks, and Ducts

The lymphatic system transports lymph from tissues back into the bloodstream.

□ Pathway of lymph flow:

Lymphatic capillaries → Collecting vessels → Lymphatic trunks → Lymphatic ducts →
Subclavian veins

◆ Lymphatic Capillaries

Lymphatic capillaries are the smallest lymphatic vessels.

Functions:

- Collect excess interstitial fluid
- Begin lymph transport

Features:

- ✓ Present in most tissues
- ✓ More permeable than blood capillaries
- ✓ Slightly larger than blood capillaries

A special type:



Lacteals

- Found in small intestine
- Absorb fats from digestion



Lymphatic Vessels

Lymphatic capillaries join together to form larger lymphatic vessels.

Characteristics:

- Thin walls
- Contain many valves
- Prevent backward movement of lymph

They are similar to veins but have:



Thinner walls



More valves



Lymphatic Trunks

As lymphatic vessels combine, they form larger trunks.

[25/06/2026 11:35 PM] hssnd: There are six major lymphatic trunks:

1 Jugular trunks

→ Drain head and neck

2 Subclavian trunks

→ Drain upper limbs

3 Bronchomediastinal trunks

→ Drain thorax

4 Intercostal trunks

→ Drain thoracic wall

5 Intestinal trunks

→ Drain digestive organs

6 Lumbar trunks

→ Drain lower limbs and lumbar region



Lymphatic Ducts

The lymphatic trunks drain into two major collecting ducts:



1. Thoracic Duct

The largest lymphatic duct.



Begins at:



Cisterna chyli

Located near the:



L2 vertebral level

It drains:



Entire body below the diaphragm



Left upper limb



Left side of head and neck



Left side of thorax

Ends by emptying into:



Left subclavian vein

2. Right Lymphatic Duct

Drains the:

✓ Right upper quadrant of the body

Includes:

- Right side of head and neck
- Right upper limb
- Right side of thorax

It empties into:

→ Right subclavian vein

Lymphatic System of Human Body

